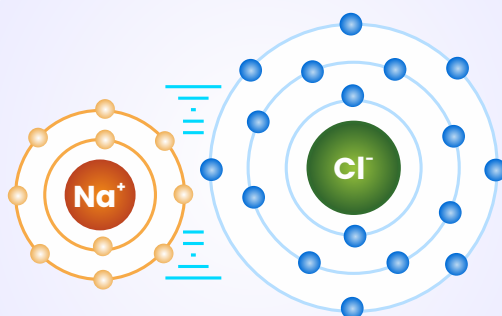


# INORGANIC CHEMISTRY

## JEE (MAIN+ADVANCED)

NURTURE COURSE



**Study Material**

**Chemical Bonding**

(English Medium)

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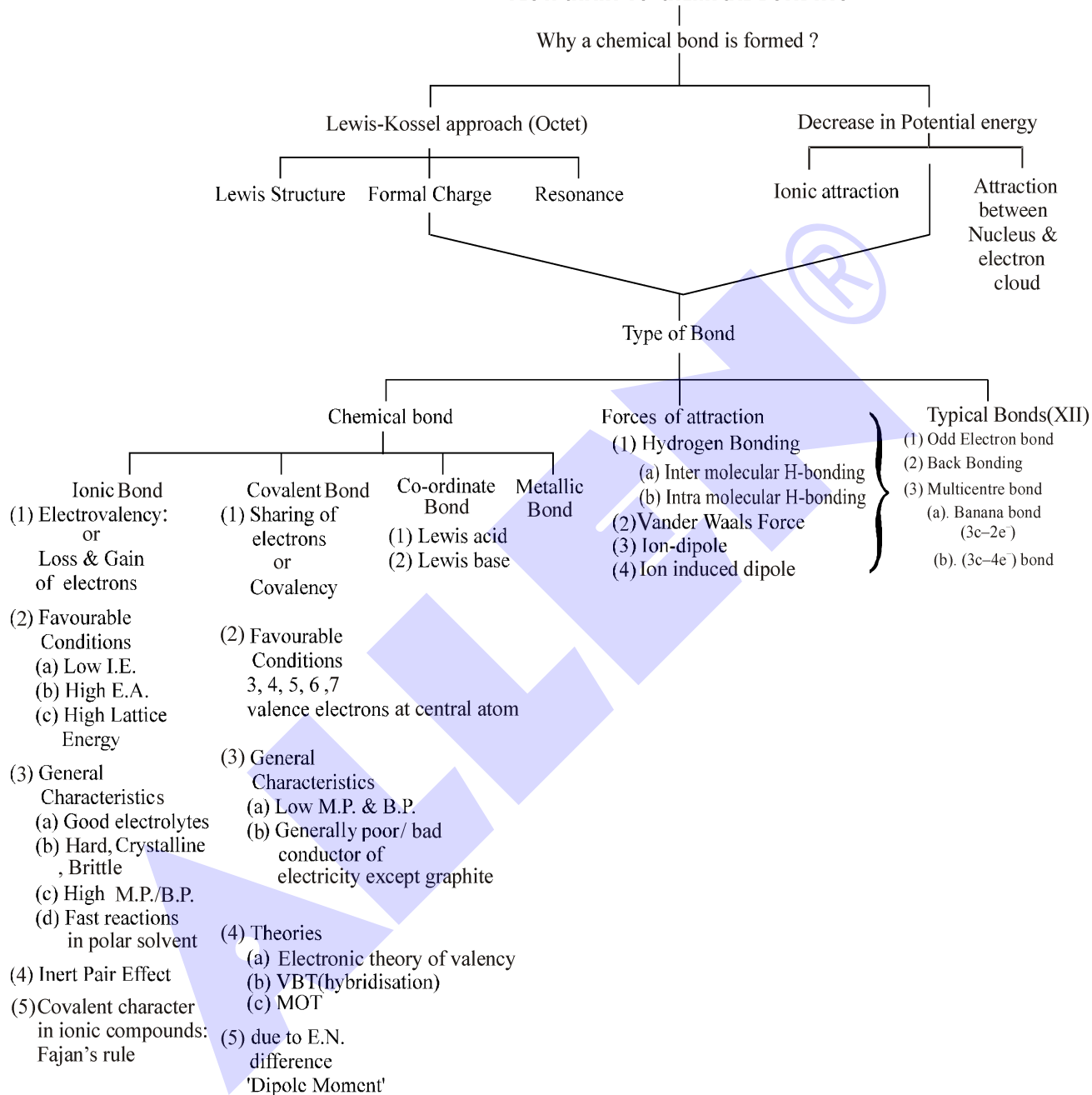
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# CHEMICAL BONDING

## FLOWCHART TO CHEMICAL BONDING



## KEY CONCEPT

### CHEMICAL BOND

- The attractive force which holds various constituents (atoms, ions, etc.) together in different chemical species is called a chemical bond.
- It is the combination of two or more chemical species involving redistribution of  $e^-$  among them.
- This process is accompanied with decrease in energy.
- Decrease in energy strengthens the bond.
- Therefore, molecules are more stable than atoms.

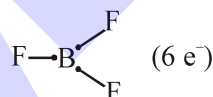
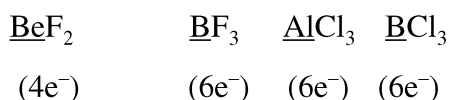
### CAUSE OF CHEMICAL COMBINATION

- Tendency to acquire minimum energy :**
- Tendency to acquire noble gas configuration (Octet rule) :**

Atoms combine to complete an octet of electrons in their outer most shell. Hence all atoms have a tendency to acquire octet ( $ns^2np^6$ ) configuration or to attain nearest noble gas configuration in their outermost orbit. This can be achieved by combining with other atom or ion.

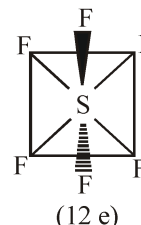
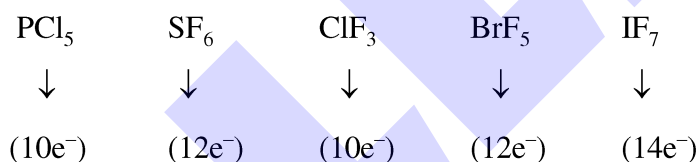
### LIMITATION OF OCTET RULE

- Contraction of octet (incomplete octet)**



These compounds are hypovalent.

- Expansion of Octet (due to empty d-orbitals)**



These compounds are hypervalent.

- Odd electron species**

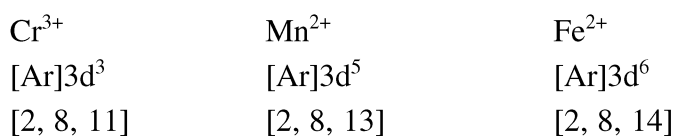
Ex. NO, NO<sub>2</sub>, ClO<sub>2</sub> etc.

### OTHER EXCEPTIONS OF OCTET RULE

- Compounds of Noble gases**

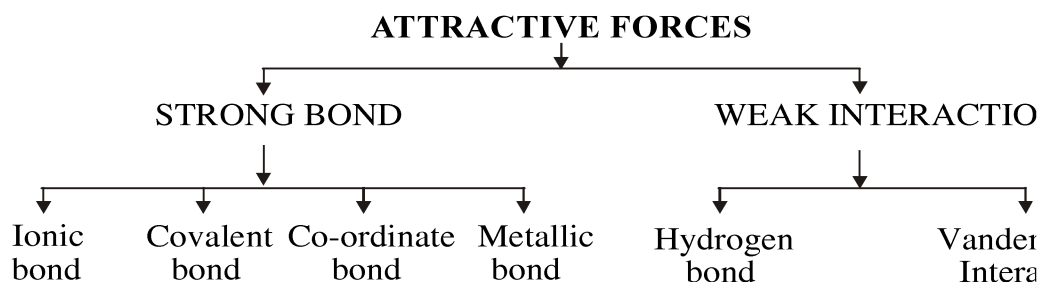
Noble gases which have already completed their octets (or duplet in case of He.) should not form compounds. However, their compounds like XeF<sub>2</sub>, XeF<sub>6</sub> & KrF<sub>2</sub> etc., have been actually prepared.

- Transition metal ions**



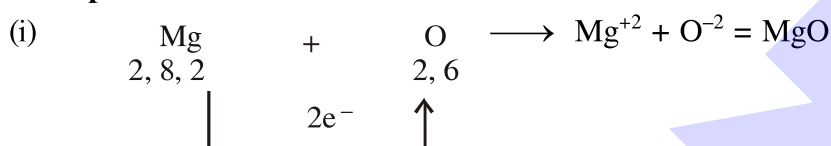
- Pseudo inert gas configuration  $[(ns^2np^6nd^{10})]$**



**CLASSIFICATION OF BONDS****ELECTROVALENT OR IONIC BOND**

(a) The chemical bond formed between two or more atoms as a result of the complete transfer of one or more electrons from one atom to another is called as Ionic or electrovalent bond.

**Example**



electrovalency of Mg = 2

electrovalency of O = 2

**Note :** Ionic bond is non-directional.

**FACTORS FAVOURING IONIC BONDING**

- Less Ionization energy of atom forming cation
- Higher electron affinity of atom forming anion
- Greater Lattice energy of formed product.
- Greater Electronegativity difference between atoms forming cation & anion.

**COVALENT BOND**

(a) A covalent bond is formed by the mutual sharing of electrons between two atoms.

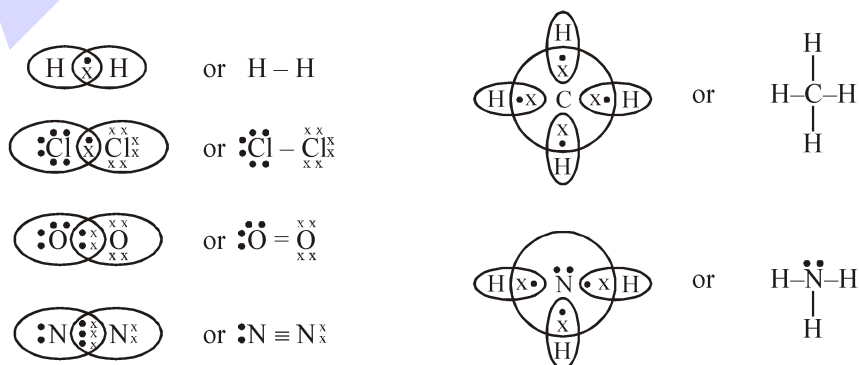


(b) The shared pair of electrons must have opposite spins, and are localised between two atoms concerned.

(c) On the basis of electrons being shared between two atoms the bonds are of three types –

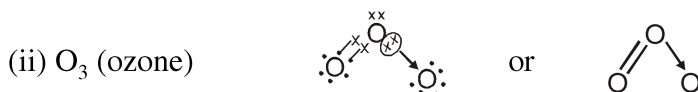
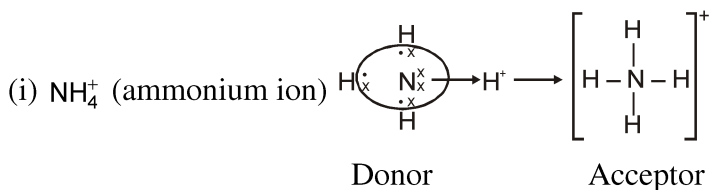
**Covalency :** Capacity to form covalent bond is known as covalency

It is represented by (–) small line. – means single bond ; = means double bond ;  $\equiv$  means triple bond.



## COORDINATE BOND (DATIVE BOND)

The bond formed between two atom in which contribution of an electron pair is made by one of them while the sharing is done by both.

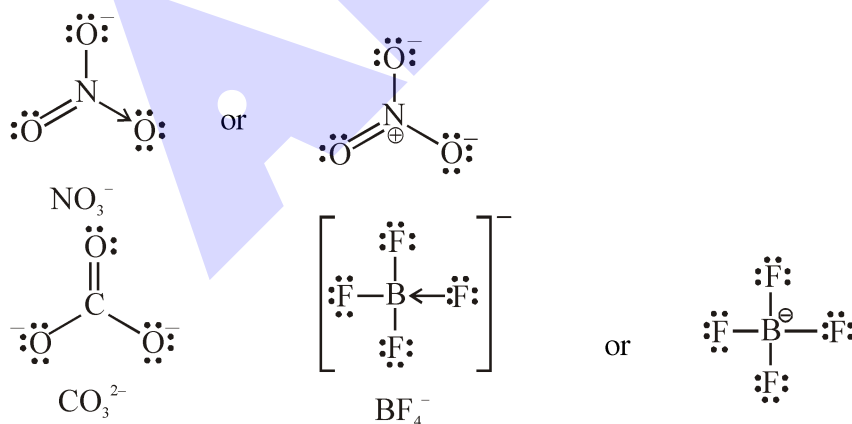


### Lewis Dot structures:

- ✦ Arrangement of various atoms in a molecule & types of bonding present in it but no idea of geometry of the molecule.
- ✦ In most cases we can construct a Lewis structure in three steps :
  - (1) Decide on the number of electrons that are included in the structure by adding together the number of all the valence electrons provided by the atoms.
  - (2) Write the chemical symbols of the atoms in the arrangement that shows which atoms are bonded together.
  - (3) Distribute the electrons in pairs so that there is one pair of electrons forming a single bond between each pair of atom bonded together and then supply electron pairs (to form lone pairs or multiple bonds) until each atom has an octet.
- ✦ In oxy acids all 'H' atoms are attached to oxygen as  $-\text{OH}$  groups except in  $\text{H}_3\text{PO}_3$  (dibasic),  $\text{H}_3\text{PO}_2$  (monobasic) &  $\text{H}_4\text{P}_2\text{O}_5$  (dibasic). Which are directly bonded to central atom.

### Applications :

- ✦ To know various linkages present
- ✦ To calculate formal charge of various elements.



## FORMAL CHARGE

Lewis dot structures, in general, do not represent the actual shapes of the molecules. In case of polyatomic ions, the net charge is possessed by the ion as a whole and not by a particular atom. It is, however, feasible to assign a formal charge on each atom.

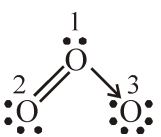
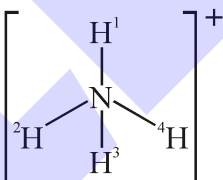
$$\text{Formal Charge : } Q_F = N_A - N_{L.P.} - \frac{1}{2} N_{B.P.}$$

Where :

$N_A$  = Total number of valence electron in the free atom

$N_{L.P.}$  = Total number of non bonding (lone pair) electrons

$N_{B.P.}$  = Total number of bonding (shared) electrons

| Molecule | Structure                                                                           | Formal Charge                                                                                                    |
|----------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| $O_3$    |    | $O(1) = 6 - 2 - \frac{1}{2} (6) = +1$ $O(2) = 6 - 4 - \frac{1}{2} (4) = 0$ $O(3) = 6 - 6 - \frac{1}{2} (2) = -1$ |
| CO       |  | $C = 4 - 2 - \frac{1}{2} \times 6 = -1$ $O = 6 - 2 - \frac{1}{2} \times 6 = +1$                                  |
| $NH_4^+$ |  | $N = 5 - 0 - \frac{1}{2} (8) = +1$ $\text{On each H} = 1 - 0 - \frac{1}{2} (2) = 0$                              |

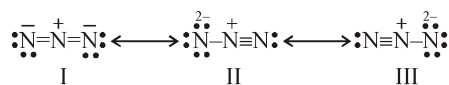
## RESONANCE\*

When a molecule cannot be completely represented by a single Lewis structure but its characteristic properties can be described by two or more different structures, with similar energy position of nuclei, bonding and non-bonding pairs of electrons, then the true structure is said to be resonance hybrid of these structures. The phenomenon is called resonance and different contributing structures are called resonating structures or canonical structures.

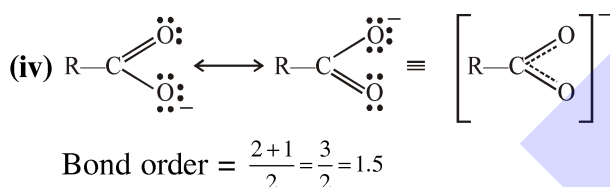
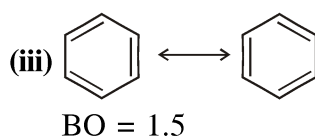
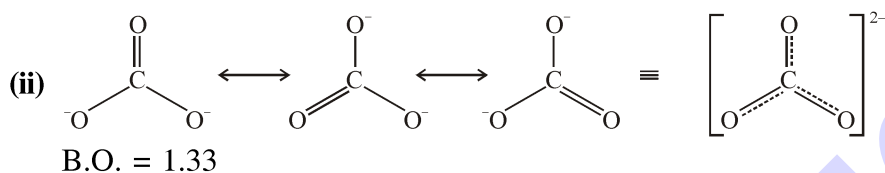
- ❖ Resonance stabilizes the molecule as the energy of the resonance hybrid is less than the energy of any single canonical structure.
- ❖ Resonance averages the bond characteristics as a whole.
- ❖ The canonical forms have no real existence.

**Resonance Structure of Some Molecules/Ions :****(i) Azide ion,  $N_3^-$  :**

The azide ion may be represented as -



The structures II and III contribute equally and the molecule has almost double bond character in each N–N bond.



\*Descriptive discussion of concept of resonance will be done in **Organic Chemistry**.

**VALENCE BOND THEORY**

**Valence bond theory** was introduced by Heitler and London (1927) and developed further by Pauling and others. A discussion of the valence bond theory is based on the knowledge of atomic orbitals, electronic configurations of elements, the overlap criteria of atomic orbitals and the hybridization of atomic orbitals.

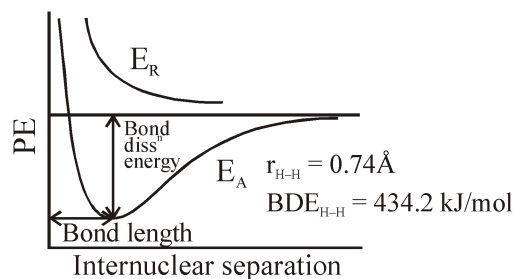
**Formation of  $H_2$  molecule :**

When two 'H'-atoms approaches towards each other for the formation of  $H_2$  molecule. The following interactions takes place.

- (i) e-e repulsion
- (ii) e-p attraction
- (iii) p-p repulsion

Experimentally, it has been found that the magnitude of attractive forces is more than the repulsive forces. As a result, when two atoms approach each other then potential energy decreases. Ultimately a stage is reached where the net force of attraction balances the force of repulsion and system acquires minimum energy.

At this stage two hydrogen atoms are said to be bonded together to form a stable molecule having the bond length of 74 pm. Since the energy gets released when the bond is formed between two hydrogen atoms, the hydrogen molecule is more stable than that of isolated hydrogen atoms.

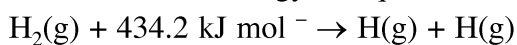


The potential energy curve for the formation of  $H_2$  molecule as a function of internuclear distance of the H atoms. The minimum in the curve corresponds to the most stable state of  $H_2$ .

**Bond Length :** Internuclear distance at minimum potential energy is called bond length.

**Bond Dissociation Energy :** The amount of energy released when one mole of same type of bonds are formed is called B.D.E.

434.2 kJ/mol of energy is required to dissociate one mole of  $H_2$  molecule.



## ORBITAL OVERLAP CONCEPT

In the formation of hydrogen molecule, there is a minimum energy state when two hydrogen atoms are so near that their atomic orbitals undergo partial interpenetration. This partial merging of atomic orbitals is called overlapping of atomic orbitals which results in the pairing of electrons. The extent of overlap decides the strength of a covalent bond. In general, greater the overlap the stronger is the bond formed between two atoms. Therefore, according to orbital overlap concept, the formation of a covalent bond between two atoms results by pairing of electrons, present in the valence shell having opposite spins.

### Directional Properties of Bonds :

The valence bond theory explains the formation and directional properties of bonds in polyatomic molecules like  $CH_4$ ,  $NH_3$  and  $H_2O$ , etc. in terms of overlap and hybridisation of atomic orbitals.

### Types of Overlapping and Nature of Covalent Bonds

The covalent bond may be classified into following types depending upon the types of overlapping :-

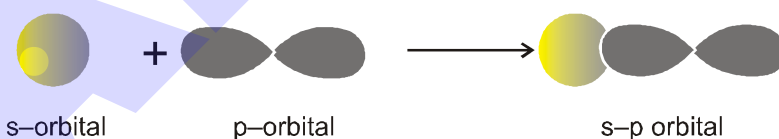
(i) sigma( $\sigma$ ) bond (ii) pi( $\pi$ ) bond (iii) delta( $\delta$ ) bond

- (i) **Sigma ( $\sigma$ ) bond :** This type of covalent bond is formed by the end to end (head on or axial) overlap of bonding orbitals along the internuclear axis. This can be formed by any one of the following types of combinations of atomic orbitals.

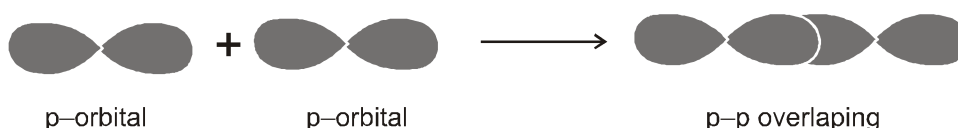
- s-s overlapping : In this case, there is overlap of two half filled s-orbitals along the internuclear axis as shown below :



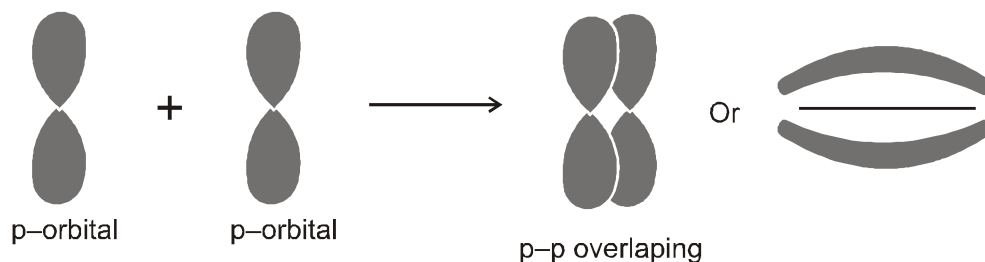
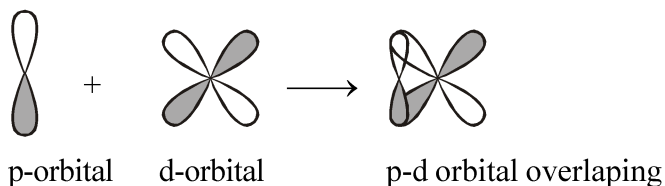
- s-p overlapping: This type of overlap occurs between half filled s-orbital of one atom and half filled p-orbital of another atom.



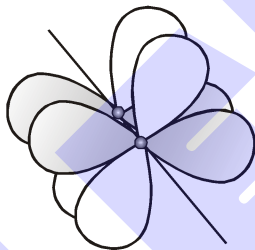
- p-p overlapping : This type of overlap takes place between half filled similar p-orbitals of the two approaching atoms.



- (ii) **pi( $\pi$ ) bond :** In the formation of  $\pi$  bond the atomic orbitals overlap in such a way that their axis remain parallel to each other and perpendicular to the internuclear axis. The orbitals formed due to sidewise overlapping consists of two saucer type charged clouds above and below the plane of the participating atoms.

(a)  $p_{\pi}-p_{\pi}$ (b)  $p_{\pi}-d_{\pi}$ 

- (iii) **delta ( $\delta$ ) bond** : are the covalent bonds where four lobes of d-orbital of one atom overlap with four lobes of the similar d-orbital of other atom. Except  $d_{z^2}$  all d orbitals form  $\delta$  bond.

**Strength of sigma and pi Bonds :**

Basically the strength of a bond depends upon the extent of overlapping- In case of sigma bond, the overlapping of orbitals takes place to a larger extent. Hence, it is stronger as compared to the pi bond where the extent of overlapping occurs to a smaller extent. Further, it is important to note that pi bond between two atoms is formed in addition to a sigma bond. It is always present in the molecules containing multiple bond (double or triple bonds)

**Advantages :**

- ❖ It explain various bond characteristics e.g., bond length, bond strength.
- ❖ It explains the quantitative relationship between the extent of overlapping and bond dissociation energy.
- ❖ This theory accounts for shape and nature of bonding of the molecule whose covalency is not according to the number of half-filled orbitals present in the ground state.
- ❖ This theory redefined the stability of molecules e.g.  $\text{BF}_3$ ,  $\text{AlCl}_3$ ,  $\text{PCl}_5$ ,  $\text{SF}_6$  etc which are exception to octet rule.

**Disadvantages :**

- ❖ According to this theory three bond angle in  $\text{CH}_4$  should be  $90^\circ$ , as these are formed by p-p overlapping, but actually it has four  $109^\circ 28'$  angles. In  $\text{NH}_3$  &  $\text{H}_2\text{O}$ , angle should be  $90^\circ$ . This is in disagreement with the actual bond angles of  $107^\circ$  &  $104.5^\circ$  in  $\text{NH}_3$  &  $\text{H}_2\text{O}$  molecules respectively.

**In order to explain the characteristic geometrical shapes of polyatomic molecules like  $\text{CH}_4$ ,  $\text{NH}_3$ ,  $\text{H}_2\text{O}$  etc. Pauling introduced the concept of hybridisation. (Given on next page)**

## HYBRIDISATION

The phenomenon of intermixing of the orbitals of slightly different energies so as to redistribute their energies, resulting in the formation of new set of orbitals of equivalent energies and shape. For example when one 2s and three 2p-orbitals of carbon hybridise, there is the formation of four new  $sp^3$  hybrid orbitals.

**Salient features of hybridisation:** The main features of hybridisation are as under :

1. The number of hybrid orbitals is equal to the number of the atomic orbitals that get hybridised.
2. The hybridised orbitals are always equivalent in energy and shape.
3. The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals.
4. These hybrid orbitals are directed in space in some preferred direction to have minimum repulsion between electron pairs and thus a stable arrangement. Therefore, the type of hybridisation indicates the geometry of the molecules.

### Important conditions for hybridisation

- (i) The orbitals present in the valence shell of the atom are hybridised.
- (ii) The orbitals undergoing hybridisation should have almost equal energy.
- (iii) Promotion of electron is not essential condition prior to hybridisation.
- (iv) It is not necessary that only half filled orbitals participate in hybridisation. In some cases, even filled orbitals of valence shell take part in hybridisation.

### Types of Hybridisation

There are various types of hybridisation involving s, p and d orbitals. The different types of hybridisation are as under :

- (i)  $sp$  hybridisation      (ii)  $sp^2$  hybridisation      (iii)  $sp^3$  hybridisation  
 (iv)  $sp^3d$  hybridisation      (v)  $sp^3d^2$  hybridisation      (vi)  $sp^3d^3$  hybridisation

### Determination of hybridisation state –

To predict hybridisation following formula may be used :

Number of hybrid orbital = Steric Number (S.N.) = Number of  $\sigma$ -bond around that atom + Number of lone pair on that atom.

| Molecule | Method             | Hybridisation         |
|----------|--------------------|-----------------------|
| $NH_4^+$ | $S.N. = 4 + 0 = 4$ | $sp^3$ hybridisation. |

| Number of hybrid orbitals | Hybridisation |
|---------------------------|---------------|
| two                       | $sp$          |
| three                     | $sp^2$        |
| four                      | $sp^3$        |
| five                      | $sp^3d$       |
| six                       | $sp^3d^2$     |
| seven                     | $sp^3d^3$     |

**Hybridisation in Ionic solid species :**

| Species                                 | Cationic part                            | Anionic part                             |
|-----------------------------------------|------------------------------------------|------------------------------------------|
| $\text{PCl}_5$                          | $\text{PCl}_4^+ (\text{sp}^3)$           | $\text{PCl}_6^- (\text{sp}^3\text{d}^2)$ |
| $\text{PBr}_5$                          | $\text{PBr}_4^+ (\text{sp}^3)$           | $\text{Br}^-$                            |
| $\text{XeF}_6$                          | $\text{XeF}_5^+ (\text{sp}^3\text{d}^2)$ | $\text{F}^-$                             |
| $\text{N}_2\text{O}_5$                  | $\text{NO}_2^+ (\text{sp})$              | $\text{NO}_3^- (\text{sp}^2)$            |
| $\text{N}_2\text{O}_3(\text{s})$        | $\text{NO}^+$                            | $\text{NO}_2^- (\text{sp}^2)$            |
| $\text{N}_2\text{O}_4(\text{s})$        | $\text{NO}^+$                            | $\text{NO}_3^- (\text{sp}^2)$            |
| $\text{I}_2\text{Cl}_6 (\text{liquid})$ | $\text{ICl}_2^+ (\text{sp}^3)$           | $\text{ICl}_4^- (\text{sp}^3\text{d}^2)$ |
| $\text{I}_2(\text{liquid})$             | $\text{I}_3^+ (\text{sp}^3)$             | $\text{I}_3^- (\text{sp}^3\text{d})$     |
| $\text{Cl}_2\text{O}_6$                 | $\text{ClO}_2^+ (\text{sp}^2)$           | $\text{ClO}_4^- (\text{sp}^3)$           |

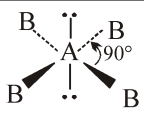
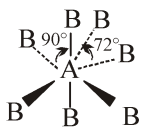
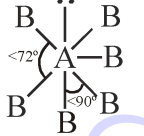
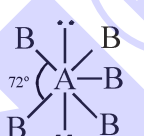
$\text{PF}_5(\text{s})$  exist in form of trigonal bipyramidal geometry.

**VALENCE SHELL ELECTRON PAIR REPULSION THEORY (VSEPR)**

- (a) Molecules having covalent bond have definite geometry as covalent bonding has directional characteristics. A simple theory was given for the molecular shape of the covalent molecules by Gillespie and Nyholm in 1957.
- (b) This theory predicts the shape of the molecule by considering the most stable configuration of the bond angles in the molecule. This theory states
  - (i) Electron pairs in the valence shell of the central atom of a molecule, whether bonding or lone pairs are regarded as occupying localised orbitals. These orbitals arrange themselves in so as to minimize the mutual electronic repulsions.
  - (ii) The magnitude of the different types of electronic repulsions follows the order given below:  
**lone pair-lone pair > lone pair - bonded pair > bonded pair - bonded pair**
  - (iii) The electronic repulsion between two pairs of electrons will be minimum if they are as far apart as possible.
  - (iv) The actual shape of the molecules containing lone pairs is a little distorted from the basic shape as in the  $\text{NH}_3$  and  $\text{H}_2\text{O}$  molecules, the bond angles are not  $109^\circ 28'$  but  $107^\circ$  and  $104.5^\circ$  respectively due to presence of one lone pair in  $\text{NH}_3$  and two lone pairs in  $\text{H}_2\text{O}$ .

SHAPES OF MOLECULES BASED ON VSEPR THEORY

| Total no. of hybrid orbitals | No. of b.p. (bond pairs) | No. of unshared pair i.e. $l_p$ | General formula | Type of hybridisation | Stereo chemical formula/str. | Shape                            | Exam.                                               |
|------------------------------|--------------------------|---------------------------------|-----------------|-----------------------|------------------------------|----------------------------------|-----------------------------------------------------|
| 2                            | 2                        | 0                               | $AB_2$          | sp                    |                              | linear                           | $BeCl_2$                                            |
| 3                            | 3                        | 0                               | $AB_3$          | $sp^2$                |                              | Trigonal planar                  | $BCl_3$ ,<br>$NO_3^-$ ,<br>$GaF_3$ ,<br>$CO_3^{2-}$ |
| 3                            | 2                        | 1                               | $AB_2$          | $sp^2$                |                              | V or Bent or angular             | $SnCl_2$ ,<br>$O_3$ , $SO_2$                        |
| 4                            | 4                        | 0                               | $AB_4$          | $sp^3$                |                              | Tetrahedron                      | $CH_4$ ,<br>$SiF_4$ ,<br>$NH_4^+$                   |
| 4                            | 3                        | 1                               | $AB_3$          | $sp^3$                |                              | Trigonal pyramidal               | $NH_3$ ,<br>$CH_3^-$                                |
| 4                            | 2                        | 2                               | $AB_2$          | $sp^3$                |                              | V or Bent or angular             | $H_2O$ ,<br>$SF_2$                                  |
| 4                            | 1                        | 3                               | $AB$            | $sp^3$                |                              | linear                           | $ClO^-$                                             |
| 5                            | 5                        | 0                               | $AB_5$          | $sp^3d$               |                              | Trigonal bipyramidal             | $PF_5$ ,<br>$SF_5^+$ ,<br>$SbBr_5$ ,<br>$XeO_3F_2$  |
| 5                            | 4                        | 1                               | $AB_4$          | $sp^3d$               |                              | Seesaw                           | $SF_4$                                              |
| 5                            | 3                        | 2                               | $AB_3$          | $sp^3d$               |                              | T-shaped                         | $ClF_3$ ,<br>$BrF_3$                                |
| 5                            | 2                        | 3                               | $AB_2$          | $sp^3d$               |                              | Linear                           | $ICl_2^-$ ,<br>$XeF_2$ ,<br>$I_3^-$                 |
| 6                            | 6                        | 0                               | $AB_6$          | $sp^3d^2$             |                              | Octahedral or Square bipyramidal | $SF_6$ ,<br>$IF_6^+$                                |
| 6                            | 5                        | 1                               | $AB_5$          | $sp^3d^2$             |                              | Square pyramidal                 | $IF_5$ ,<br>$XeOF_4$ ,<br>$BrF_5$                   |

|   |   |   |        |           |                                                                                    |                        |                              |
|---|---|---|--------|-----------|------------------------------------------------------------------------------------|------------------------|------------------------------|
| 6 | 4 | 2 | $AB_4$ | $sp^3d^2$ |  | Square planar          | $IF_4^-$ , $XeF_4$ , $ICl_4$ |
| 7 | 7 | 0 | $AB_7$ | $sp^3d^3$ |  | Pentagonal bipyramidal | $IF_7$                       |
| 7 | 6 | 1 | $AB_6$ | $sp^3d^3$ |  | Distorted octahedral   | $XeF_6$ , $IF_6^-$           |
| 7 | 5 | 2 | $AB_5$ | $sp^3d^3$ |  | Pentagonal planar      | $XeF_5^-$                    |

## BOND PARAMETERS

### (I) Bond order

### (III) Bond Angle

### (II) Bond Length (Bond distance)

### (IV) Bond Enthalpy

#### (I) Bond order:

The Bond Order is given by the number of bonds between the two atoms in a molecule.

The bond order, for example in  $H_2$  (with a single shared electron pair), in  $O_2$  (with two shared electron pairs) and in  $N_2$  (with three shared electron pairs) is 1, 2, 3 respectively.

**Note :** Isoelectronic molecules and ions have identical bond orders; for example,  $F_2$  and  $O_2^{2-}$  have bond order 1.  $N_2$ ,  $CO$  and  $NO^+$  have bond order 3.

**Note :** A general correlation useful for understanding the stabilities of molecules is that: with increase in bond order, bond enthalpy increases and bond length decreases.

#### (II) Bond Length:

Bond length is defined as the equilibrium distance between the nuclei of two bonded atoms in a molecule. Bond lengths are measured by spectroscopic, X-ray diffraction and electron-diffraction techniques about which you will learn in higher classes.

#### Factors affecting bond length

- |                             |                                             |
|-----------------------------|---------------------------------------------|
| (a) Size of atoms           | (b) Effect of bond order or number of bonds |
| (c) Effect of Resonance     | (d) Effect of Electronegativity difference  |
| (e) Effect of Hybridisation |                                             |

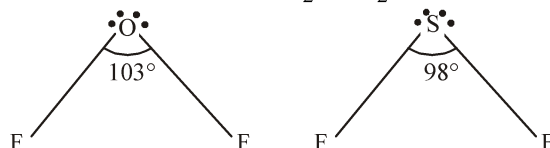
#### (III) Bond angle:

It is defined as the angle between the orbitals containing bonding electron pairs around the central atom in a molecule/complex ion. Bond angle is expressed in degree which can be experimentally determined by spectroscopic methods. For example  $H-\hat{O}-H$  bond angle in water can be represented as under :



**Comparison of bond angles.**

- (a) If central atoms are in different hybridisations then it can be compared.  
 (b) If same hybridisation but different central atoms then bond angle would be more of the molecule in which central atom is more E.N. eg.  $\text{OF}_2$ ,  $\text{SF}_2$



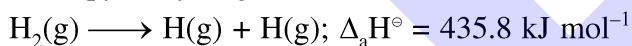
- (c) Molecules having same C.A. but different substituent then bond angle increases as the size of attached atom increases except in symmetrical molecules and  $\text{H}_2\text{O}$  /  $\text{F}_2\text{O}$ ,  $\text{NH}_3$  /  $\text{NF}_3$

**Factors affecting bond angle**

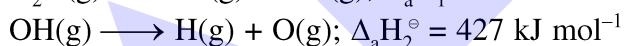
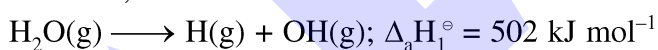
- (a) State of Hybridisation  
 (b) Presence of lone pair  
 (c) Electronegativity of central atom  
 (d) Electronegativity of surrounding atom  
 (e) Size of surrounding atom  
 (f) Lone pairs may sometimes be transferred from a filled shell of one atom to an unfilled shell of another bonded atom, causing less repulsion.  
 (g) Multiple bond orbital repel other orbitals more strongly than single bond orbitals.

**(IV) Bond Enthalpy**

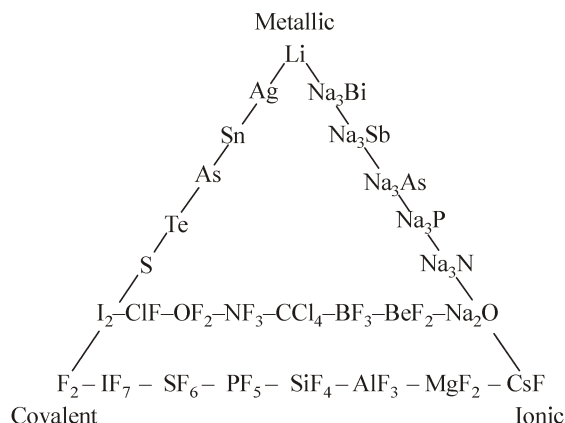
It is defined as the amount of energy required to break one mole of bonds of a particular type between two atoms in a gaseous state. The unit of bond enthalpy is  $\text{kJ mol}^{-1}$ . For example, the H – H bond enthalpy in hydrogen molecule is  $435.8 \text{ kJ mol}^{-1}$ .



⇒ In polyatomic molecules the **term mean or average bond enthalpy** is used. It is obtained by dividing total bond dissociation enthalpy by the number of bonds broken as explained below in case of water molecule,



$$\text{Average bond enthalpy} = \frac{502 + 427}{2} = 464.5 \text{ kJ mol}^{-1}$$

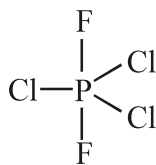
**SUMMARY OF THE THREE MAIN TYPES OF BONDS**

**BENT'S RULE**

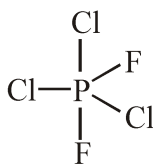
- (i) A lone pair of electron prefers to occupy that hybrid orbitals which has greater percentage of s-character.
- (ii) A more electronegative atom/group prefers to occupy that hybrid orbital which has smaller percentage of s-character.

**Ex.** Draw the geometry of  $\text{PCl}_3\text{F}_2$

**Sol.**



Correct  
Structure



Wrong  
Structure

Because highly electronegative atom occupy axial position (axial position has smaller percentage of s-character).

**DRAGO'S RULE**

On the basis of experimental bond angles of certain molecules fulfilling the following three conditions,

- (i) Central atom belongs to third or lower period in periodic table
- (ii) Central atom must contain atleast one lone pair of electron, and
- (iii) Electronegativity of surrounding atom is  $\leq 2.5$

Drago generalised that in such molecules justification of experimental bond angle can be made satisfactorily if one considers no hybridisation, i.e., overlapping of almost pure atomic orbitals from central atom.

In such molecules bond angle is approximately  $90^\circ$ .

| Group 15       | Bond angle      | Group 16              | Bond angle      |
|----------------|-----------------|-----------------------|-----------------|
| $\text{NH}_3$  | $107^\circ 48'$ | $\text{H}_2\text{O}$  | $104^\circ 28'$ |
| $\text{PH}_3$  | $93^\circ 36'$  | $\text{H}_2\text{S}$  | $92^\circ$      |
| $\text{AsH}_3$ | $91^\circ 48'$  | $\text{H}_2\text{Se}$ | $91^\circ$      |
| $\text{SbH}_3$ | $91^\circ 18'$  | $\text{H}_2\text{Te}$ | $90.5^\circ$    |

- Right order of bond angle.

(a)  $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$

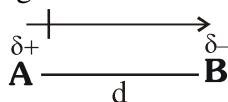
(b)  $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$

**MOLECULES THAT DO NOT EXIST**

- (1)  $\text{SF}_4$ ,  $\text{SF}_6$  &  $\text{PF}_5$  exist while,  $\text{OF}_4$ ,  $\text{OF}_6$ ,  $\text{NF}_5$  do not exist
- (2) (a)  $\text{PI}_5(\text{vap})$  &  $\text{SbCl}_6$  do not exist
- (b)  $\text{SbCl}_6$  does not exist while  $\text{TeCl}_6$  exist
- (c)  $\text{PI}_5$  (Solid) exist
- (3)  $\text{SF}_6$ ,  $\text{PF}_5$ ,  $\text{XeF}_6$ ,  $\text{XeF}_4$  &  $\text{XeF}_2$  exist while  $\text{SH}_6$ ,  $\text{PH}_5$ ,  $\text{XeH}_6$ ,  $\text{XeH}_4$ ,  $\text{XeH}_2$  do not exist

## DIPOLE MOMENT

- The degree of polarity of covalent bond is given by the dipole moment ( $\mu$ )
- It is the product of the either charge and the distance between them.



Electronegativity of A < Electronegativity of B

$$\mu = \delta \times d$$

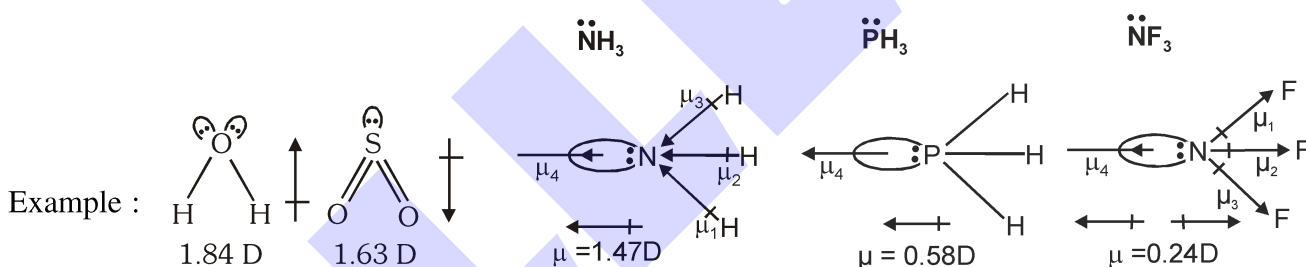
- Dipole moment is a vector quantity.  
The direction of dipole moment is represented by  $\rightarrow$
- Units = Cm (S.I.) or esu cm (CGS) or Debye (common unit)
- 1 D =  $10^{-18}$  esu cm =  $3.33 \times 10^{-30}$  coulomb m

### Dipole moment depends on

- Electronegativity difference between bonded atoms
- Direction of bond dipole moment
- Angle between various bonds
- Influence of unshared  $e^-$  pairs
- Magnitude of polarity of the molecule
- Symmetrical / Unsymmetrical shape.

### Application of dipole moment

- To determine the polarity and geometry of molecules



- To calculate the percentage of ionic character

$$\% \text{ ionic character} = \frac{\text{observed } \mu}{\text{calculated } \mu \text{ (for 100 \% ionic)}} \times 100 \%$$

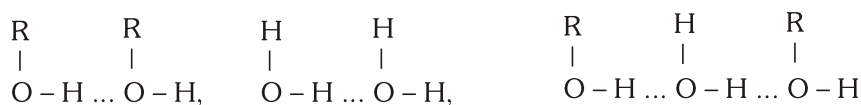
## HYDROGEN BONDING

- Hydrogen bonding :** When a hydrogen atom is bonded to a highly electronegative atom (like F, O or N) comes under the influence of another strongly electronegative atom, then a weak electrostatic force of attraction is developed between them, which is called as hydrogen bond.

### Types of H-bonding:

- Intermolecular H-bond

- This type of H-bonding takes place between two molecules. **Ex.** ROH,  $H_2O$ , R - OH &  $H_2O$

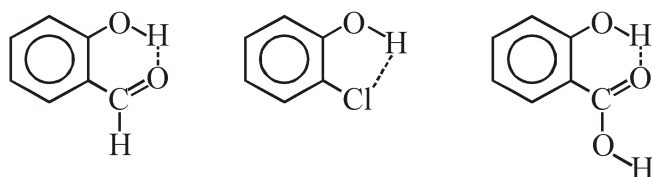


- In such compounds molecular wt., M.P. & B.P. are high.

- Extent of Inter molecular H-bonding  $\uparrow$  viscosity & density  $\uparrow$ .

★ Intramolecular H-bond

Normally when 2 hydroxyl groups are present on the same carbon atom i.e. gem diols are unstable, but "chloralhydrate" is a stable molecule due to formation of H-bond.



**Properties influenced by hydrogen bonding**

- Abnormal behaviour of water.
- Association of molecules eg. dimerisation of  $\text{CH}_3\text{COOH}$ ,  $\text{HCOOH}$
- Dissociation of a polar species.
- Abnormal melting point & boiling point.
- Enhanced solubility in water.

★ **Metallic bonds :**

Electron gas model or sea model, with metal atom existing as kernels along with less firmly held valence electrons & bonds between various kernels (at the lattice site) & valence electrons are known as metallic bonds.

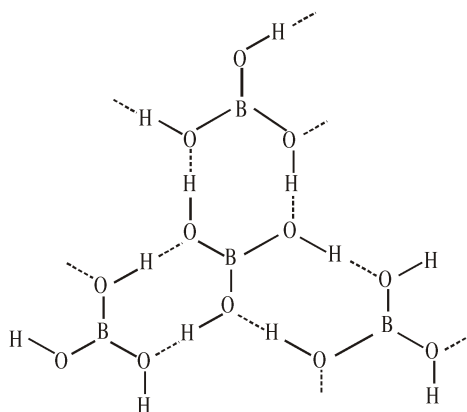
**QUESTION OF HYDROGEN BONDING**

- Q.1 Explain the structure of Boric acid in solid state.
- Q.2 Boiling point of o-Nitrophenol is less than meta and para nitrophenol. Why?
- Q.3 Maleic acid is more acidic than fumaric acid. Why?
- Q.4  $\text{H}-\text{F}$  is only liquid among halogen acid. Why?
- Q.5 Ammonia is more easily liquefied than  $\text{HCl}$ , explain.
- Q.6 Why ice floats on water?
- Q.7 Water shows maximum density at  $4^\circ\text{C}$ . Why?
- Q.8  $\text{HI}$  is the strongest halogen acid, whereas  $\text{H}-\text{F}$  is the weakest. Why?
- Q.9 Wood pieces are used to hold ice-cream. Why?
- Q.10  $\text{KHF}_2$  is possible but not  $\text{KBr}_2$  or  $\text{KI}_2$ . Why?
- Q.11 o-Nitrophenol is less soluble in  $\text{H}_2\text{O}$  than p-Nitrophenol. Why?
- Q.12 o-Hydroxy benzaldehyde is a liquid at room temperature while p-hydroxy benzaldehyde is a high melting solid.
- Q.13 Glycerol is more viscous than ethanol. Explain.
- Q.14  $\text{CH}_4$  and  $\text{H}_2\text{O}$  have nearly same molecular weight. Yet  $\text{CH}_4$  has a boiling point 112 K and water 373 K. Explain.

- Q.15 The experimental molecular weight of acetic acid is just double than theoretical molecular weight of acetic acid. Why?
- Q.16 Although chlorine has same electronegativity as nitrogen but the former does not form effective H-bonding. Explain.
- Q.17 Molar entropy change of vapourization of acetic acid is less than that of water. Explain
- Q.18 Heat of vapourization of water is higher than HF, however strength of H-bond in HF is higher than water. Explain

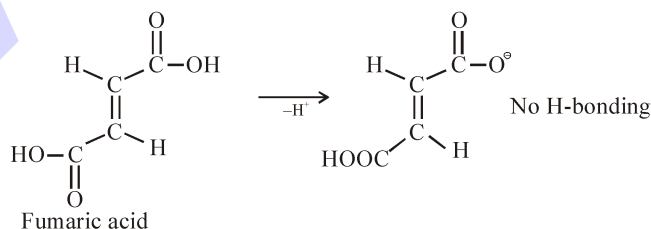
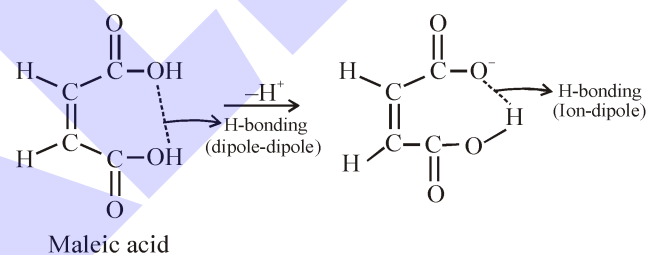
### ANSWER OF HYDROGEN BONDING

1.

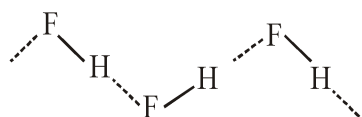


Structure of boric acid; the dotted lines represent hydrogen bonds

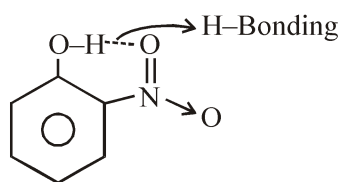
2. Inter molecular H-Bonding in case of para nitro phenol is strong enough to increase boiling point which is not favoured in o-Nitro phenol.
3. Maleic acid anion gets stabilize due to intra molecular H-bonding which is not possible in anion of Fumaric acid. So  $H^+$  releasing tendency is more in case of maleic acid



4. Due to strong H-Bonding in H-F

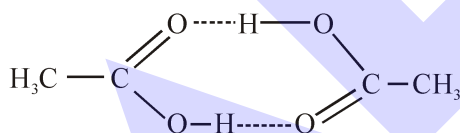


5. Due to H-Bonding in  $\text{NH}_3$ .
6. Density of ice is lower than that of water due to cage like structure of ice volume of ice increases and density decreases.
7. On heating ice in temperature range  $0-4^\circ\text{C}$  H-bonds of ice break down, which decreases volume, hence density of  $\text{H}_2\text{O}$  increases. On heating ice after  $4^\circ\text{C}$  thermal vibrations of  $\text{H}_2\text{O}$  molecules increases which increases effective volume of ice, so density of ice again decreases.
8. H-I bond is weak as compare to H-F so it can be dissociated easily and can give  $\text{H}^+$  easily.
9. The organic matter in wood participate in H-bonding with water molecules in ice-cream.
10.  $\text{K}^+[\text{F}-\text{H}\cdots\text{F}]$  But  $\text{K}\text{HBr}_2$  &  $\text{K}\text{HI}_2$  can't form H-Bond.  
 $\downarrow$   
 H-bonding
11. In o-Nitrophenol intra molecular H-bond is present which decreases it's solubility in water.

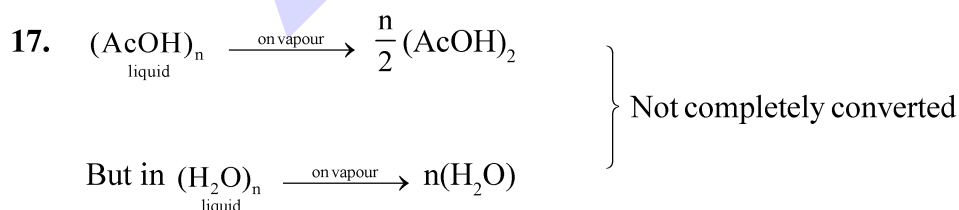


Ortho Nitro-phenol

12. In o-hydroxy benzaldehyde intramolecular H-bond is present which is less stronger than intermolecular H-bond in p-hydroxy benzaldehyde.
13. Extent of H-bonding in glycerol is more as compare to ethanol so it is more viscous.
14. Due to H-bonding in  $\text{H}_2\text{O}$
15. Due to H-bonding in acetic acid it forms dimer so overall molecular weight is just double of original

Dimer of  $\text{CH}_3\text{COOH}$ 

16. Due to bigger size of 'Cl' atom it's interaction is not enough to evolve the amount of energy which lies in the range of H-bond.



Here all  $\text{H}_2\text{O}$  molecule gets vapourised. So entropy change is more

18. As extent of H-bond is more in  $\text{H}_2\text{O}$  as compare to HF, heat of vaporisation of water is higher than HF.

**VANDER WAAL'S FORCES**

- ⇒ These are the weakest type of inter molecular forces that exist among the chemical species which bring about a significant change in physical properties.
- ⇒ These are non-directional, non-valence cohesive forces. These attractive forces being played between the two molecules are independent of the presence of other molecules.
- ⇒ Solid, liquid or gaseous states of many molecules are explained on the basis of inter molecular forces other than covalent, ionic or metallic bonds. Although inert gases do not form any type of bond but may exist in liquid and solid states. This shows that the atoms of inert gases are attracted by each other through some type of inter molecular forces. These intermolecular forces are called Vander Waals forces.

**Types of Vander Waal's Forces**

- (1) **Dipole-dipole interaction (Keesom forces)** : The force of attraction between the oppositely charged poles of two polar molecules (for example :  $\text{H}_2\text{S}$ ,  $\text{HCl}$ ,  $\text{PH}_3$  etc.) is called dipole-dipole attraction.
- (2) **Dipole-induced dipole interaction (Debye forces)** : This type of cohesive forces occurs in a mixture of polar and non polar molecules. The former induce polarity in non polar molecules by disturbing their electron system. for example force of attraction between  $\text{Cl}_2$  and  $\text{H}_2\text{O}$ .
- (3) **Instantaneous dipole-Induced dipole interaction (Dispersion forces/London forces)** : The weak intermolecular forces operating in similar non polar gaseous molecules are called London forces. These forces are very weak in nature and exist only at low temperature. For example weak intermolecular forces in  $\text{F}_2$ ,  $\text{Cl}_2$ ,  $\text{N}_2$ , molecules and in noble gases.

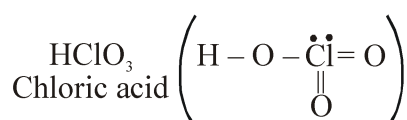
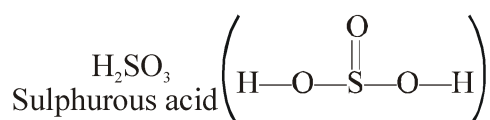
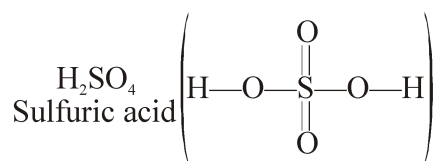
(Note :- London forces present in both polar and non polar species)

**Other Weak Interactions**

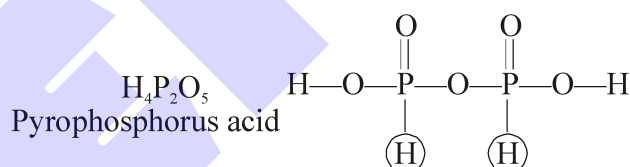
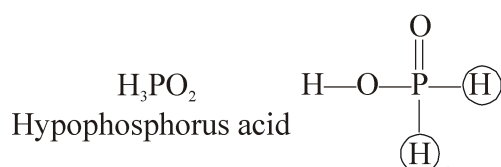
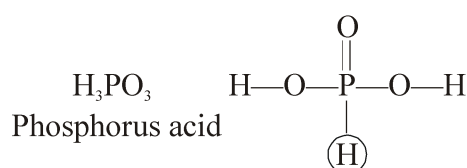
- (1) **Ion-dipole interaction** : Polar molecules are attracted by ions. The negative pole is attracted by cation and positive pole attracted by the anion. This type of attraction is called ion dipole attraction, ion-dipole attraction is observed generally in the process of solvation when sodium chloride ( $\text{Na}^+ \text{Cl}^-$ ) is dissolved in water because negative poles of water aggregate around  $\text{Na}^+$  ions and positive poles around  $\text{Cl}^-$  ions.
- (2) **Ion-induced dipole interaction** : When non polar molecules come in contact with ions, its electron cloud gets polarised and the oppositely charged end of it is attracted by the ion. For example attraction between  $\text{I}^-$  and  $\text{I}_2$  molecule.

## ❑ OXY-ACIDS

Those compounds which contain X–O–H bond are called oxy-acids. Where X is usually a non-metal. But sometimes X may also be a metal.

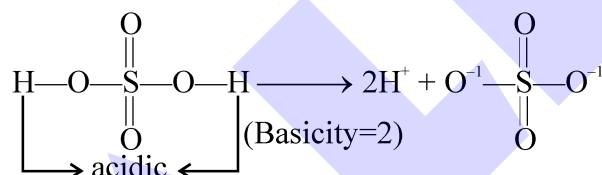


In oxy acids, all the hydrogens are attached with oxygen. Although there are a very few exceptions (as given below), in which all hydrogens are not attached with oxygen.

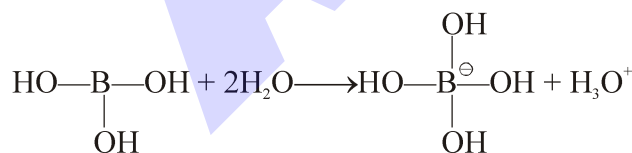


In these three oxyacids circled hydrogens are not attached with oxygen.

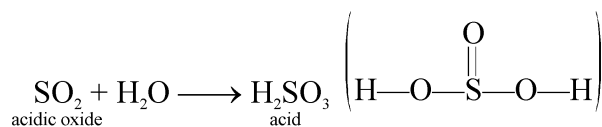
In oxyacids, those hydrogens which are attached with oxygen are acidic in nature, i.e., they ionise in solution.

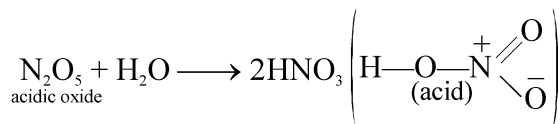
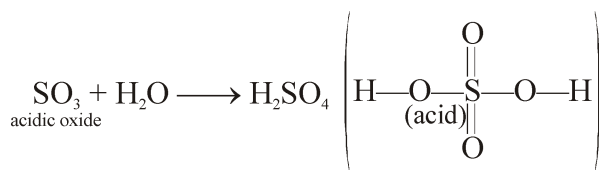


No. of hydrogens attached with oxygen is known as basicity of the acid. One notable exception is  $\text{H}_3\text{BO}_3$  in which all the three hydrogens are attached with oxygen yet, they do not ionise in solution. In fact,  $\text{H}_3\text{BO}_3$  when dissolved in water, it behaves as a monobasic acid.



Oxyacids are actually aqueous solutions of acidic oxides.



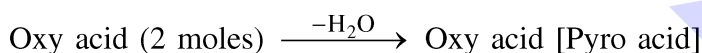


- Some acids have suffix-ic acid and some has-ous acid. Usually, when the central atom in oxy acid is in higher oxidation state the name of the acid ends with -ic acid and when the central atom is in lower oxidation state, its name ends with -ous acid.

**Oxyacids can be classified as**

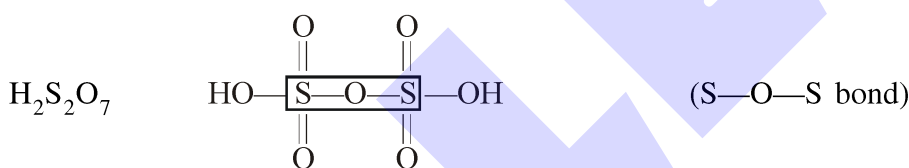
### □ PYRO ACIDS :

When one mole of water is removed from two moles of an oxyacid, the resultant oxy acid is known as pyro acid. In general pyro acids contain X—O—X bonds.



- $\text{H}_2\text{SO}_4$  (2 moles) =  $\text{H}_4\text{S}_2\text{O}_8 \xrightarrow{-\text{H}_2\text{O}} \text{H}_2\text{S}_2\text{O}_7$  [Pyrosulphuric acid]  
[Sulphuric acid]

In general Pyro acids contain X—O—X bond.



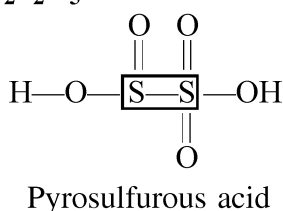
- $\text{H}_3\text{PO}_4$  (2 moles) =  $\text{H}_6\text{P}_2\text{O}_8 \xrightarrow{-\text{H}_2\text{O}} \text{H}_4\text{P}_2\text{O}_7$  [Pyrophosphoric acid]  
[Orthophosphoric acid]



- $2\text{H}_3\text{PO}_3$  (phosphorus acid)  $\xrightarrow{-\text{H}_2\text{O}} \text{H}_4\text{P}_2\text{O}_5$  (pyrophosphorus acid)  $\text{HO}-\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \boxed{\text{P}-\text{O}-\text{P}} \\ \parallel \quad \parallel \\ \text{H} \quad \text{H} \end{array}-\text{OH}$

- $2\text{H}_2\text{SO}_3 \xrightarrow{-\text{H}_2\text{O}} \text{H}_2\text{S}_2\text{O}_5$  (Pyrosulfurous acid)

**Exception :  $\text{H}_2\text{S}_2\text{O}_5$  [Pyrosulfurous acid] does not contain S—O—S linkage.**



## ❑ HYPO ACID :

Hypo prefix is added to those oxy acids which is obtained when one oxygen atom is removed from ous form of (usually) acids.

Oxy acid(-ous form)  $\xrightarrow{-O}$  Oxy acid (Hypo -ous acid) [It is a trick to understand hypo formation]

- $H_3PO_3$  [Phosphorous acid] – O  $\longrightarrow$   $H_3PO_2$  [Hypophosphorous acid] monobasic acid

- $\begin{matrix} +3 \\ HXO_2 \\ \text{Halousacid} \\ (x=Cl, Br, I) \end{matrix} \xrightarrow{-[O]} \begin{matrix} +1 \\ HOX \\ \text{Hypohalous acid} \end{matrix}$

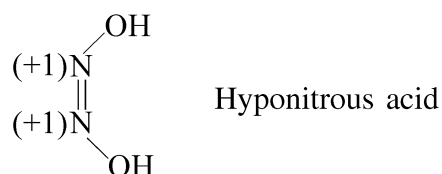
HOCl - Hypochlorous acid

HOBr - Hypobromous acid

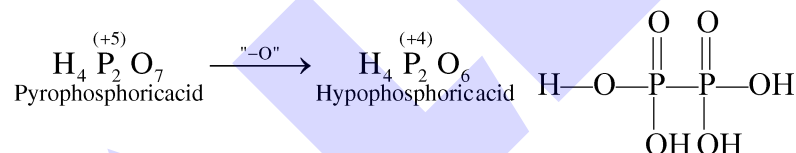
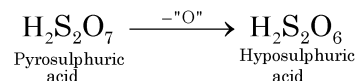
HOI - Hypoiodous acid

- $\begin{matrix} +3 \\ HNO_2 \\ \text{Nitrous acid} \end{matrix} \xrightarrow{-[O]} \begin{matrix} +1 \\ HNO \end{matrix}$

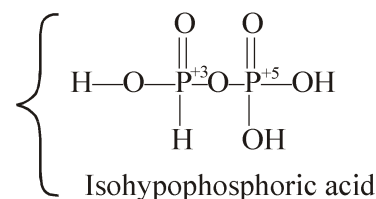
HNO exist in dimeric form i.e. as  $(HNO)_2 = H_2N_2O_2$



**If one O is removed from pyro 'ic' acid it produce hypo 'ic' acid**



Exception : (It may be exist as isohypophosphoric acid)



Isohypophosphoric acid

## ❑ PER ACID AND PEROXY ACID :

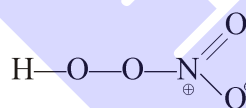
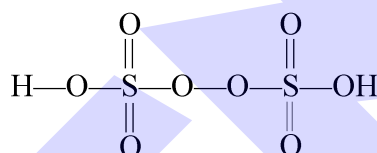
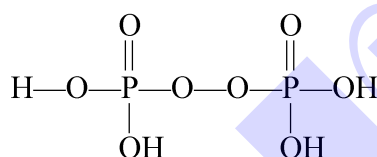
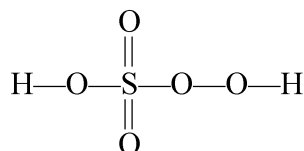
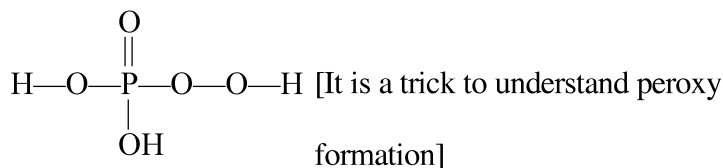
When one oxygen is added to -ic form of acid and if the oxidation number of central atom in resulting oxy acid is more than the -ic form of acid, then it is per acid.

- $\begin{matrix} (+5) \\ HXO_3 \\ \text{Halic acid} \end{matrix} + O \longrightarrow \begin{matrix} (+7) \\ HXO_4 \\ \text{Perhalic acid} \end{matrix} \quad \begin{array}{c} O \\ || \\ H-O-X=O \\ || \\ O \end{array}$

$HClO_4$  perchloric acid  $HBrO_4$  - Perbromic acid.

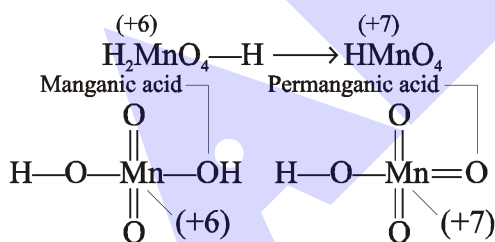
**Peroxy prefix** is added to those oxy acids which are obtained when one oxygen is added to –ic from of acid and the oxidation number of central atom remains the same. Peroxy acids contain peroxy linkage ( $\text{—O—O—}$ ).

- $$\begin{array}{ccc} +5 & & +5 \\ \text{H}_3\text{PO}_4 + \text{O} & \longrightarrow & \text{H}_3\text{PO}_5 \\ \text{Orthophosphoric} & & \text{Peroxyphosphoric} \\ \text{acid} & & \text{acid} \end{array}$$
- $$\begin{array}{ccc} +6 & & +6 \\ \text{H}_2\text{SO}_4 + \text{O} & \longrightarrow & \text{H}_2\text{SO}_5 \\ \text{Pyrosulphuric} & & \text{Peroxydisulphuric} \\ \text{disulphuric acid} & & \text{acid (Caro's acid)} \end{array}$$
- $$\begin{array}{ccc} +5 & & +5 \\ \text{H}_4\text{P}_2\text{O}_7 + \text{O} & \longrightarrow & \text{H}_4\text{P}_2\text{O}_8 \\ \text{diphosphoric acid} & & \text{Peroxy diphosphoric} \\ \text{pyrophosphoric acid} & & \text{acid} \end{array}$$
- $$\begin{array}{ccc} +6 & & +6 \\ \text{H}_2\text{S}_2\text{O}_7 + \text{O} & \longrightarrow & \text{H}_2\text{S}_2\text{O}_8 \\ \text{Pyrosulphuric acid} & & \text{Peroxydisulphuric} \\ \text{disulphuric acid} & & \text{acid (Marshall acid)} \end{array}$$
- $$\begin{array}{ccc} +5 & & +5 \\ \text{HNO}_3 + \text{O} & \longrightarrow & \text{HNO}_4 \\ \text{(nitric acid)} & & \text{(peroxy nitric acid)} \end{array}$$



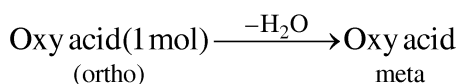
**Note :** Peroxynitric acid does not contain  $\text{N—O—H}$  bond, yet it is considered to be oxy acid.

**Exception :**  $\text{H}_2\text{MnO}_4$  is manganic acid when one hydrogen is removed from  $\text{H}_2\text{MnO}_4$ ,  $\text{HMnO}_4$  is formed which is named as permanganic acid.



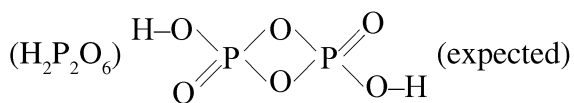
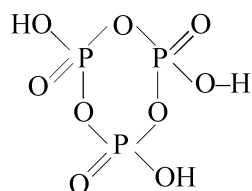
## □ ORTHO ACIDS AND META ACIDS :-

When one mole of  $\text{H}_2\text{O}$  is removed from one mole of an oxy acid, the resulting compound, if an oxy acid (i.e. it contains at least one  $\text{X—OH}$  bond) then the oxy acid from which  $\text{H}_2\text{O}$  is removed is named as ortho acid and the product oxyacid is known as META ACID.

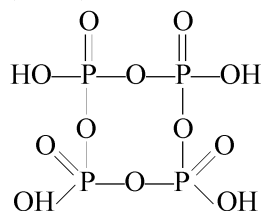


- $$\text{H}_3\text{PO}_4 \xrightarrow{-\text{H}_2\text{O}} \text{HPO}_3 \text{ Meta phosphoric acid (It forms Cyclic polymetaphosphoric acids)}$$

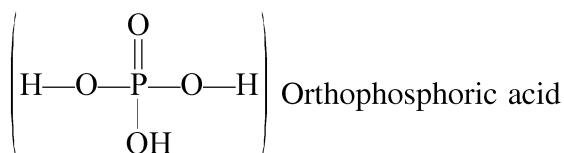
(ortho – phosphoric acid)

♦ **Cyclic polymetaphosphoric acids :**(HPO<sub>3</sub>)<sub>2</sub> - Dimetaphosphoric acid(HPO<sub>3</sub>)<sub>3</sub>

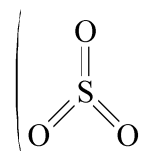
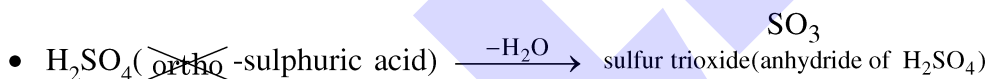
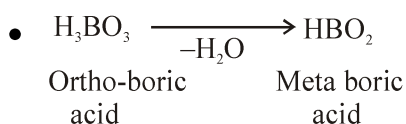
Trimetaphosphoric acid

(HPO<sub>3</sub>)<sub>4</sub>

Tetrametaphosphoric acid



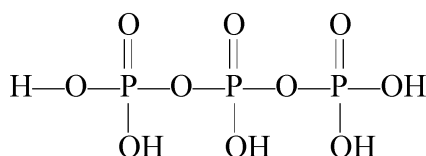
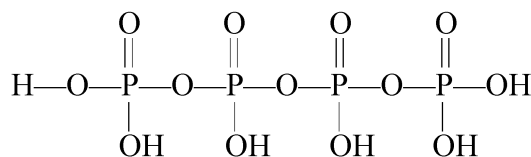
An ortho acid must contain at least three hydrogens.



As SO<sub>3</sub> does not contain any hydrogen (i.e., X-OH/S-OH bond), so SO<sub>3</sub> is not an oxy acid rather it is anhydride of sulphuric acid (H<sub>2</sub>SO<sub>4</sub>) so, H<sub>2</sub>SO<sub>4</sub> can't be prefixed with ortho, and its name does not contain any prefix and it is named as sulphuric acid.

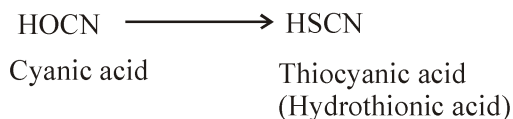
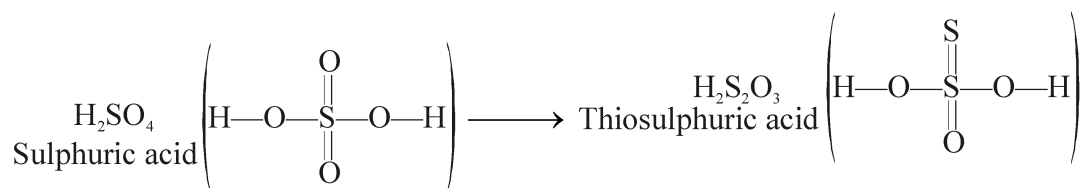
□ **POLYPHOSPHORIC ACID :**

Oxy acids having general formula H<sub>(n+2)</sub>P<sub>n</sub>O<sub>(3n+1)</sub> are known as polyphosphoric acids. These contain (n-1) P-O-P bonds.

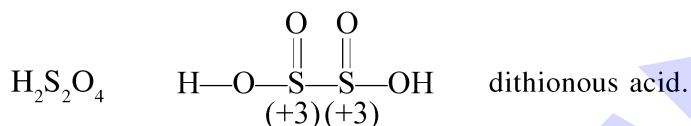
H<sub>5</sub>P<sub>3</sub>O<sub>10</sub> - Tripolyphosphoric acidH<sub>6</sub>P<sub>4</sub>O<sub>13</sub> - Tetrapolyphosphoric acid

❑ **THIO ACID :**

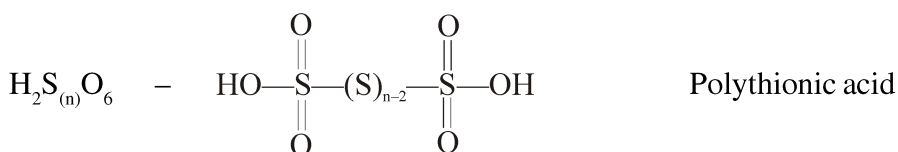
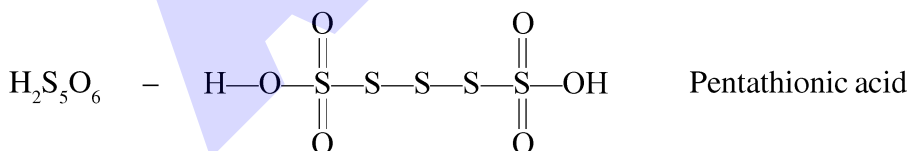
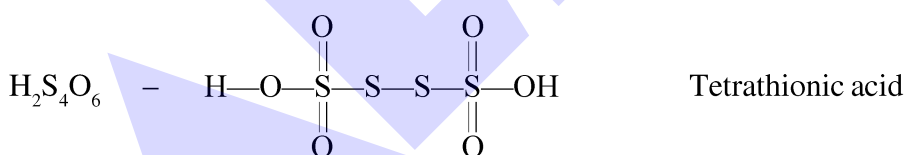
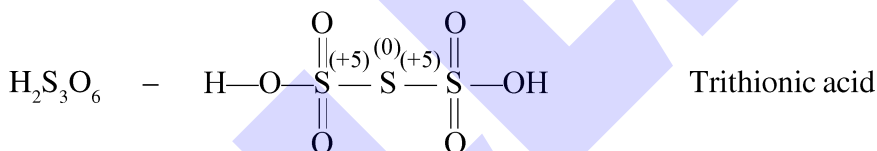
- ♦ 1. Oxy acids in which one oxygen atom are replaced by sulphur are considered as thio acid.  
For example :-



- ♦ 2. Oxy acids of sulphur containing S—S bonds are known as thio acids.



**Thionic acids :** Oxy acids of sulphur containing S—S bond and having general formula  $\text{H}_2\text{S}_{(n+2)}\text{O}_6$  are known as thionic acids



❑ TABLE OF OXIDATION STATE OF OXY-ACIDS :

| Element            | -Ous acid                                                                                                                                                                                                                                                                                                                                                                                | -ic acid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P                  | <p>(+3, <math>\text{H}_3\text{PO}_3</math>, <math>\text{H}_4\text{P}_2\text{O}_5</math>)</p> <p>(+2, <math>\text{H}_3\text{PO}_2</math>)</p>                                                                                                                                                                                                                                             | <p>(+4) <math>\text{H}_4\text{P}_2\text{O}_6</math> (Hypophosphoric acid)</p> <p>+5 <math>\begin{cases} \rightarrow \text{H}_{n+2}\text{P}_n\text{O}_{3n+1} - \text{polyphosphoric acid} \\ \rightarrow \text{H}_3\text{PO}_4, \text{H}_4\text{P}_2\text{O}_7, \text{H}_4\text{P}_2\text{O}_8 \\ \rightarrow \text{H}_3\text{PO}_5 - \text{peroxymonophosphoric acid} \end{cases}</math></p>                                                                                                           |
| S                  | <p>(+4) <math>\text{H}_2\text{SO}_3</math> Sulfurous acid</p> $\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{H}-\text{O}-\text{S}-\text{S}-\text{OH} \\ (+3) \quad (+5) \end{array}$ <p>Pyrosulfurous acid</p>                                                                                                                                          | <p>(+6) Sulphuric acid (<math>\text{H}_2\text{SO}_4</math>)</p> <p><math>\text{H}_2\text{S}_2\text{O}_7</math> - Pyrosulphuric acid</p> <p><math>\text{H}_2\text{S}_{(n+2)}\text{O}_6</math> - Thionic acids</p> <p>e.g : <math>\begin{array}{c} \text{O} \quad \text{O} \quad (+5) \\ \parallel \quad \parallel \\ \text{H}-\text{O}-\text{S}-\text{S}-\text{OH} \\ \parallel \quad \parallel \\ \text{O} \quad \text{O} \end{array}</math></p> <p>Dithionic acid</p>                                 |
|                    | <p><math>\text{H}_2\text{S}_2\text{O}_4</math> <math>\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{H}-\text{O}-\text{S}-\text{S}-\text{OH} \\ (+3) \quad (+3) \end{array}</math></p> <p>dithionous acid</p>                                                                                                                                             | <p><math>\begin{array}{c} \text{O} \quad (+5) \quad \text{O} \quad (+5) \\ \parallel \quad \parallel \quad \parallel \quad \parallel \\ \text{H}-\text{O}-\text{S}-\text{S}-\text{S}-\text{S}-\text{OH} \\ \parallel \quad (0) \quad (0) \quad \parallel \\ \text{O} \quad \text{O} \end{array}</math></p> <p>Tetrathionic acid</p> <p>(+6) <math>\text{H}_2\text{SO}_5</math> - Peroxymonosulfuric acid</p> <p>(+6) <math>\text{H}_2\text{S}_2\text{O}_8</math> - Peroxydisulfuric acid</p>           |
| X<br>(X=Cl, Br, I) | <p>(+1) <math>\text{H}-\text{O}-\text{X}</math> Hypohalous acid</p> <p>e.g. <math>\text{H}-\text{O}-\text{Cl}</math> - Hypochlorous acid</p> <p><math>\text{H}-\text{O}-\text{I}</math> - Hypoiodous acid</p> <p>(+3) <math>\text{HXO}_2</math> ; <math>\text{H}-\text{O}-\text{X}=\text{O}</math> Halous acid</p> <p><math>\text{H}-\text{O}-\text{I}=\text{O}</math> - Iodous acid</p> | <p>(+5) Halic acid (<math>\text{HXO}_3</math>)</p> $\begin{array}{c} \text{O} \\ \parallel \\ \text{H}-\text{O}-\text{X}=\text{O} \\ (+5) \end{array}$ $\begin{array}{c} \text{O} \\ \parallel \\ \text{H}-\text{O}-\text{Cl}=\text{O} \quad [\text{Chloric acid}] \end{array}$ <p>(+7) Perhalic acid (<math>\text{HXO}_4</math>)</p> $\begin{array}{c} \text{O} \\ \parallel \\ \text{e.g. } \text{H}-\text{O}-\text{Br}=\text{O} \quad [\text{Perbromic acid}] \\ \parallel \\ \text{O} \end{array}$ |
| Others             |                                                                                                                                                                                                                                                                                                                                                                                          | <p>(+3) <math>\text{H}_3\text{BO}_3</math> Orthoboric acid</p> <p>(+3) <math>\text{HBO}_2</math> - Metaboric acid</p> <p>(+4) Orthosilicic acid <math>\text{H}_4\text{SiO}_4</math></p> <p><math>\text{H}_2\text{SiO}_3</math> - Metasilicic acid</p> <p><math>\text{H}_6\text{Si}_2\text{O}_7</math> - Pyrosilicic acid</p>                                                                                                                                                                           |

- ♦ **Naming of salts of oxy acids : The name of salt ends with – “ate” if the name of the oxy acid ends with - ic acid and all prefix remain same eg. :-**

|                                                                                            |                                                                                                                                                               |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $H_2SO_4$<br>Sulphuric acid                                                                | $Na_2SO_4$ - Sodium sulfate<br>$K_2SO_4$ - Potassium sulfate<br>$BaSO_4$ - Barium sulfate<br>$FeSO_4$ - Ferrous sulfate<br>$Al_2(SO_4)_3$ - Aluminium sulfate |
| $H_3PO_4$<br>Orthophosphoric acid                                                          | $Na_3PO_4$ - Sodium orthophosphate<br>$K_3PO_4$ - Potassium orthophosphate<br>$Ca_3(PO_4)_2$ - Calcium Orthophosphate<br>$AlPO_4$ - Aluminium orthophosphate  |
| $HPO_3$<br>meta phosphoric acid                                                            | $NaPO_3$ - Sodium metaphosphate<br>$KPO_3$ - Potassium metaphosphate<br>$Ca(PO_3)_2$ - Calcium metaphosphate<br>$Al(PO_3)_3$ - Aluminium metaphosphate        |
| $HClO_3$<br>chloric acid                                                                   | $KClO_3$ - Potassium chlorate<br>$NaClO_3$ - Sodium chlorate<br>$Ca(ClO_3)_2$ - Calcium chlorate                                                              |
| $HClO_4$<br>Perchloric acid                                                                | $NaClO_4$ - Sodium perchlorate<br>$KClO_4$ - Potassium perchlorate<br>$NH_4ClO_4$ - Ammonium perchlorate<br>$Ca(ClO_4)_2$ - Calcium perchlorate               |
| $H_3PO_4$<br>Orthophosphoric acid<br>$(HPO_3)_6 = H_6P_6O_{18}$<br>Hexametaphosphoric acid | $KH_2PO_4$ potassium dihydrogen orthophosphate<br>$Ca(H_2PO_4)_2$ calcium dihydrogen orthophosphate<br>$Na_6P_6O_{18}$<br>Sodium hexametaphosphate            |
| $HBO_2$<br>metaboric acid                                                                  | $NaBO_2$ sodium metaborate                                                                                                                                    |
| $H_2SO_5$<br>Peroxisulphuric acid                                                          | $Na_2SO_5$ - Sodium peroxysulfate                                                                                                                             |
| $H_4P_2O_7$<br>Pyrophosphoric acid                                                         | $Na_4P_2O_7$ - Sodium pyrophosphate                                                                                                                           |

**If the name of the oxy acid ends with ous acid then the name of the salt ends with “ite”**

|                                                 |                                                                                               |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------|
| $HClO_2$<br>chlorous acid                       | $NaClO_2$ sodium chlorite<br>$KClO_2$ potassium chlorite<br>$Ca(ClO_2)_2$ calcium chlorite    |
| $H_3PO_3$<br>Phosphorous acid<br>(dibasic acid) | $Na_2HPO_3$ sodium phosphite<br>$K_2HPO_3$ Potassium phosphite<br>$CaHPO_3$ calcium phosphite |
| $H_2SO_3$<br>sulfurous acid                     | $Na_2SO_3$ sodium sulfite<br>$K_2SO_3$ potassium sulfite                                      |
| $HNO_2$<br>Nitrous acid                         | $NaNO_2$ sodium nitrite<br>$KNO_2$ potassium nitrite                                          |
| $HOCl$                                          | $NaOCl$ sodium hypochlorite                                                                   |

Hypochlorous acid

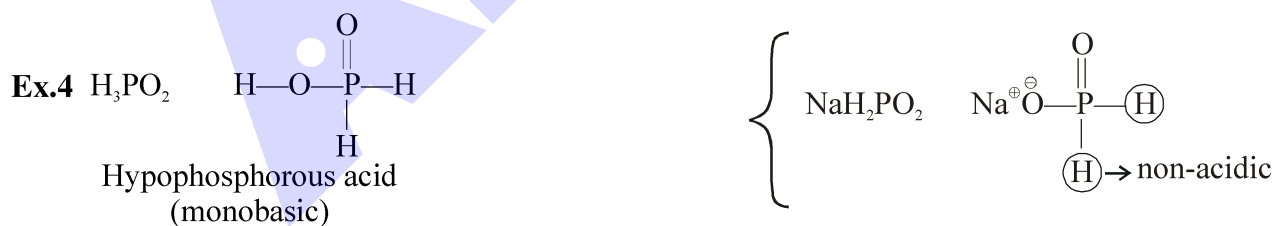
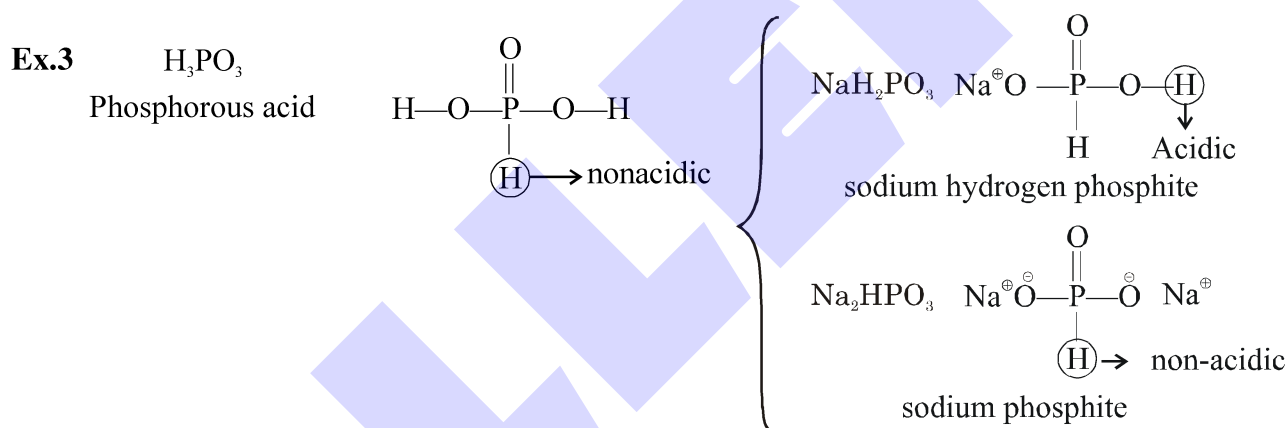
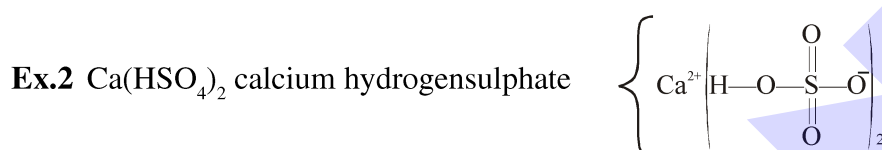
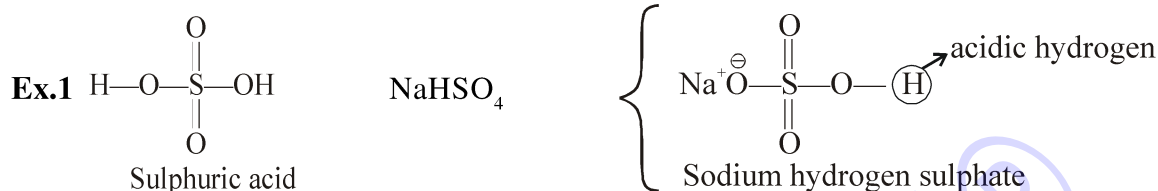
KBr potassium hypobromite



NaOI sodium hypoiodite

Hypophosphorous acid  
(monobasic acid) $\text{NaH}_2\text{PO}_2$  sodium hypophosphite $\text{Ba}(\text{H}_2\text{PO}_2)_2$  Barium hypophosphite

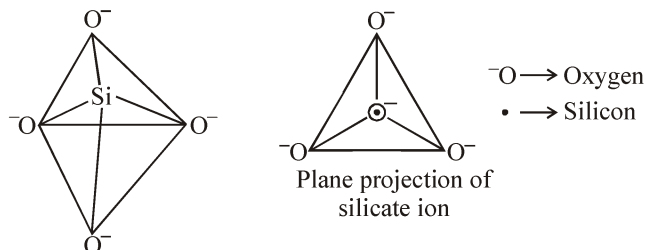
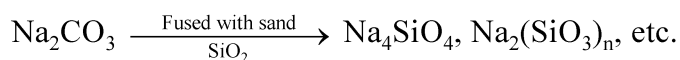
*If all the ionisable hydrogens are not removed from the acid to form salt, then the hydrogen word is added to the name of the salt containing acidic hydrogens (H attached with oxygen)*



sodium hypophosphite but not sodium dihydrogen phosphite as H is not acidic.

## SILICATES

Silicates are metal derivatives of silicic acid,  $\text{H}_4\text{SiO}_4$  or  $\text{Si}(\text{OH})_4$ . Silicates are formed by heating metal oxide or metal carbonates with sand, e.g.,



Silicates have basic unit of  $\text{SiO}_4^{4-}$ , each silicon atom is bonded with four oxide ions tetrahedrally.

There are following types of silicates

| Silicates                | Sharing of O-atom / Basic Tetrahedral unit | Contribution of O-atom / Basic Tetrahedral unit | General formula                      |
|--------------------------|--------------------------------------------|-------------------------------------------------|--------------------------------------|
| Ortho                    | 0                                          | 4                                               | $\text{SiO}_4^{4-}$                  |
| Pyro                     | 1                                          | 3.5                                             | $\text{Si}_2\text{O}_7^{6-}$         |
| Cyclic                   | 2                                          | 3                                               | $(\text{SiO}_3)_n^{2n-}$             |
| Simple chain (pyroxene)  | 2                                          | 3                                               | $(\text{SiO}_3)_n^{2n-}$             |
| Double chain (Amphibole) | (3, 2) av = 2.5                            | $\frac{11}{4} = \left(\frac{5.5}{2}\right)$     | $(\text{Si}_4\text{O}_{11})_n^{6n-}$ |
| 2D or (Sheet)            | 3                                          | 2.5                                             | $(\text{Si}_2\text{O}_5)_n^{2n-}$    |
| 3D                       | 4                                          | 2                                               | $(\text{SiO}_2)_n$                   |

Feldspar :  $\text{K}[\text{AlSi}_3\text{O}_8]$

Ultramarine :  $\text{Na}_2[(\text{AlSiO}_4)_6]\text{S}_2$

Zeolite :  $\text{Na}_2[\text{Al}_2\text{Si}_3\text{O}_{10}]\cdot 2\text{H}_2\text{O}$

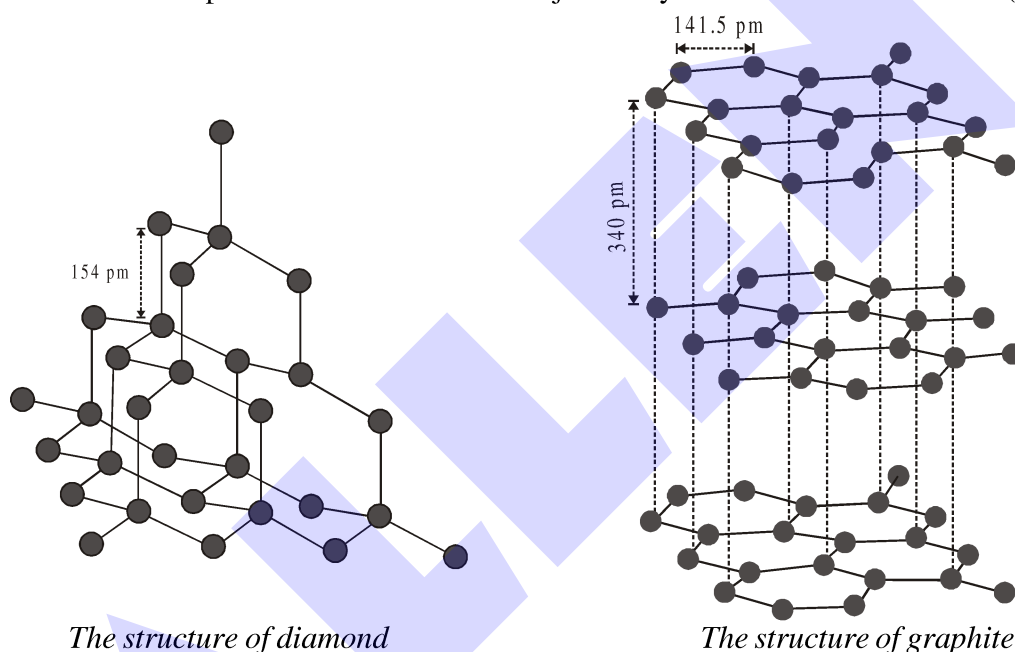
## ALLOTROPE OF CARBON FAMILY

⇒ **Diamond** (*kinetically most stable allotrope of carbon, meta stable phase of carbon*)

- It has a **crystalline lattice**. A rigid three dimensional network of carbon atoms.
- Each carbon atom undergoes  **$sp^3$  hybridisation** and linked to four other carbon atoms by using hybridised orbitals in tetrahedral fashion.
- **C—C bond length is 154 pm.**
- It is very difficult to break **extended covalent bonding** and, therefore, diamond is a **very hard** substance.

**Uses :-**

- Used as an abrasive for sharpening hard tools, in making dyes and in the manufacture of tungsten filaments for electric light bulbs.
- Diamond is a precious stone and used in jewellery. It is measured in **carats** (1 carat = 200 mg).



The structure of diamond

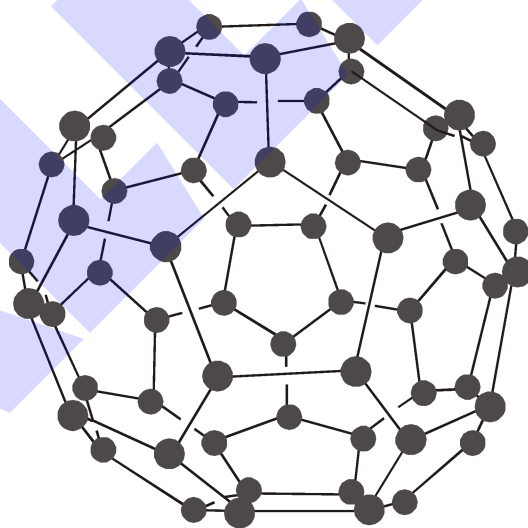
The structure of graphite

⇒ **Graphite** (*Thermodynamically most stable allotrope of carbon*)

- Layered structure. **Layers are held by van der Waals forces** and distance between two layers is **340 pm**.
- Each layer is composed of **planar hexagonal rings** of carbon atoms. C—C bond length within the layer is **141.5 pm**.
- Each carbon atom in hexagonal ring **undergoes  $sp^2$  hybridisation** and makes three sigma bonds with three neighbouring carbon atoms. **Fourth electron forms a  $\pi$  bond**. The electrons are delocalised over the whole sheet. Thus, graphite is lustrous.
- Electrons are mobile and, therefore, graphite conducts electricity along the sheet.
- Graphite cleaves easily between the layers and, therefore, it is very soft and slippery.
- **Uses :-** Due to slippery nature graphite is used as a **dry lubricant** in machines running at high temperature, where oil cannot be used as a lubricant.
- **Being good conductor, graphite is used for electrodes in batteries and industrial electrolysis.**

**Fullerenes**

- Fullerenes are made by the heating of graphite in an electric arc in the presence of inert gases such as helium or argon.
- The sooty material formed by condensation of vapourised  $C_n$  small molecules consists of mainly  $C_{60}$  with smaller quantity of  $C_{70}$  and traces of fullerenes consisting of even number of carbon atoms up to 350 or above.
- **Fullerenes are the only pure form of carbon because they have smooth structure without having 'dangling' bonds. Fullerenes are cage like molecules.**
- **Fullerene  $C_{60}$  :- molecule has a shape like soccer ball and called Buckminsterfullerene. It contains 20, six- membered rings and 12, five membered rings.**
- This ball shaped molecule has **60 vertices** and each one is occupied by one carbon atom and it also **contains both single and double bonds** with C–C distances of 143.5 pm and 138.3 pm respectively. **Spherical fullerenes are also called *bucky balls* in short.**
- A six membered ring is fused with six or five membered rings but a five membered ring can only fuse with six membered rings.
- All the carbon atoms are equal and they undergo  $sp^2$  hybridisation. Each carbon atom forms three sigma bonds with other three carbon atoms.
- The remaining electron at each carbon is delocalised in molecular orbitals, which in turn give aromatic character to molecule. However, because of non-planar nature of fullerenes, their aromatic character gets diminished and reactivity increases.



[The structure of  $C_{60}$ , Buckminsterfullerene : Note that molecule has the shape of a soccer ball (football)]

**IMPORTANT POINTS FOR CARBON**

- **Thermodynamic stability order** : Graphite > Diamond > Fullerene  $C_{60}$
- It is very important to know that **graphite is thermodynamically most stable allotrope of carbon** and, therefore,  $\Delta H_f$  of graphite is taken as zero.  $\Delta H_f$  values of diamond and fullerene,  $C_{60}$  are **1.90 and 38.1 kJ mol<sup>-1</sup>**, respectively.
- Other forms of elemental carbon like carbon black, coke, and charcoal are all impure forms of graphite or fullerenes.
  - ⇒ **Carbon black** is obtained by burning hydrocarbons in a limited supply of air.
  - ⇒ **Charcoal and coke** are obtained by heating wood or coal respectively at high temperatures in the absence of air.

**Uses of Carbon**

- **Graphite fibres** embedded in plastic material form high strength, lightweight composites. The composites are used in products such as tennis rackets, fishing rods, aircrafts and canoes.
- **Crucibles** made from graphite are inert to dilute acids and alkalies.
- Being **highly porous, activated charcoal** is used in adsorbing poisonous gases; also used in water filters to remove organic contaminants and in airconditioning system to control odour.
- **Carbon black** is used as black pigment in black ink and as filler in automobile tyres.
- **Coke** is used as a fuel and largely as a reducing agent in metallurgy.

**ALLOTROPE OF OXYGEN FAMILY**

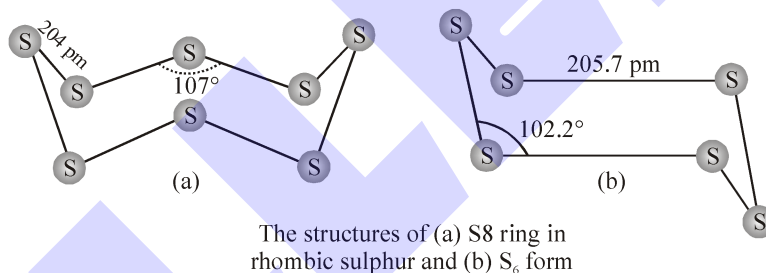
- **Oxygen**  
Two non-metallic forms, dioxygen  $O_2$  and ozone  $O_3$ . Dioxygen  $O_2$  is stable as a diatomic molecule.  $O_3$  molecule is diamagnetic while  $O_2$  is paramagnetic.
- Six allotropes of **selenium** are known.
- **Tellurium** has only one crystalline form, which is silvery white and semimetallic.
- This is similar to grey Se, but has stronger metallic interaction.
- **Sulphur Allotropic forms** :
  - Sulphur forms numerous allotropes of which the **yellow rhombic** ( $\alpha$ -sulphur) and **monoclinic** ( $\beta$ -sulphur) forms are the most important.
  - The **stable form** at room temperature is **rhombic sulphur**, which transforms to monoclinic sulphur when heated above 369 K.

⇒ **Rhombic sulphur ( $\alpha$ -sulphur)** (*Most stable sulphur allotrope at room temperature*)

- This allotrope is **yellow in colour, m.p. 385.8 K and specific gravity 2.06**. Rhombic sulphur crystals are formed on evaporating the solution of roll sulphur in  $\text{CS}_2$ .
- It is **insoluble in water** but dissolves to some extent in benzene, alcohol and ether. It is **readily soluble** in  $\text{CS}_2$ .

⇒ **Monoclinic sulphur ( $\beta$ -sulphur)**

- Its m.p. is 393 K and specific gravity 1.98. It is **soluble in  $\text{CS}_2$  while water insoluble**.
- This form of sulphur is prepared by melting rhombic sulphur in a dish and cooling, till crust is formed. Two holes are made in the crust and the remaining liquid poured out. On removing the crust, **colourless needle shaped crystals of  $\beta$ -sulphur are formed**.
- **It is stable above 369 K and transforms into  $\alpha$ -sulphur below it**. Conversely,  $\alpha$ -sulphur is stable below 369 K and transforms into  $\beta$ -sulphur above this. At 369 K, both the forms are stable. This temperature is called **transition temperature**.
- **Both rhombic and monoclinic sulphur have  $\text{S}_8$  molecules**. These  $\text{S}_8$  molecules are packed to give different crystal structures. **The  $\text{S}_8$  ring in both the forms is puckered and has a crown shape**. The molecular dimensions are given in the figure.



### EXTRA POINTS

- Several other modifications of sulphur containing 6-20 sulphur atoms per ring have been synthesised in the last two decades.
- In cyclo- $\text{S}_6$ , the ring adopts the chair form and the molecular dimensions are as shown in figure.
- At elevated temperatures ( $\sim 1000$  K),  $\text{S}_2$  is the dominant species and is paramagnetic just like  $\text{O}_2$ .
- **Engel's sulphur ( $\epsilon$ -sulphur)** is unstable and contains  $\text{S}_6$  rings arranged in the chair conformation. It is made by pouring  $\text{Na}_2\text{S}_2\text{O}_3$  solution into concentrated  $\text{HCl}$  and extracting the S with toluene.
- Plastic sulphur-( $\chi$ ) is obtained by pouring liquid sulphur into water.

### Allotropes of Nitrogen Family :

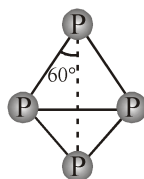
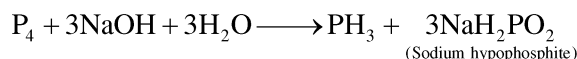
Solid As, Sb and Bi each exist in several allotropic forms. Arsenic vapour contains tetrahedral  $\text{As}_4$  molecules. A reactive yellow form of the solid resembles white phosphorus and is thought to contain tetrahedral  $\text{As}_4$  units. Sb also has a yellow form. All three elements have much less reactive metallic or  $\alpha$ -forms.

## PHOSPHORUS ALLOTROPIC FORMS

Phosphorus is found in many allotropic forms, the important ones being white, red and black.

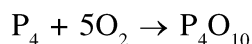
### ⇒ White phosphorus

- Translucent, white, waxy solid.
- It is poisonous, **insoluble in water** but **soluble in carbon disulphide** and glows in dark (chemiluminescence).
- It dissolves in boiling NaOH solution in an inert atmosphere giving  $\text{PH}_3$ .



Discrete tetrahedral  $\text{P}_4$  unit of White Phosphorus

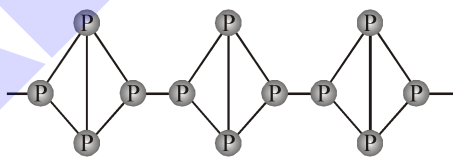
- White phosphorus is **less stable** and therefore, **more reactive** than the other solid phases under normal conditions **because of angular strain** in the  $\text{P}_4$  molecule where the angles are only  $60^\circ$ . It readily catches fire in air to give dense white fumes of  $\text{P}_4\text{O}_{10}$ .



### ⇒ Red phosphorus

- Obtained by heating white phosphorus at 573K in an inert atmosphere for several days.
- When red phosphorus is heated under high pressure, a series of phases of black phosphorus is formed.
- Red phosphorus possesses **iron grey lustre**. It is **odourless, nonpoisonous and insoluble** in water as **well as in carbon disulphide**.
- Chemically, red phosphorus is **much less reactive than white phosphorus**. It does not glow in the dark.

It is polymeric, consisting of chains of  $\text{P}_4$  tetrahedra linked together in the manner as shown in Fig.

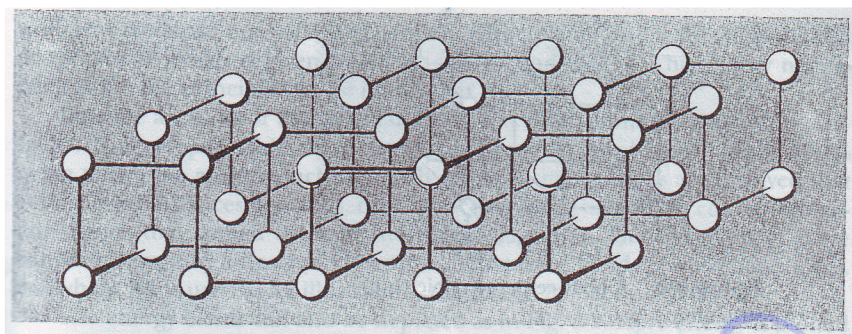


Red Phosphorus

### ⇒ Black phosphorus

- Two forms  $\alpha$ -black phosphorus and  $\beta$ -black phosphorus.
- $\alpha$ -Black phosphorus is formed when red phosphorus is heated in a sealed tube at 803K. It can be sublimed in air and has opaque monoclinic or rhombohedral crystals. It does not oxidise in air.

- $\beta$ -Black phosphorus is prepared by heating white phosphorus at 473 K under high pressure. It does not burn in air upto 673 K.

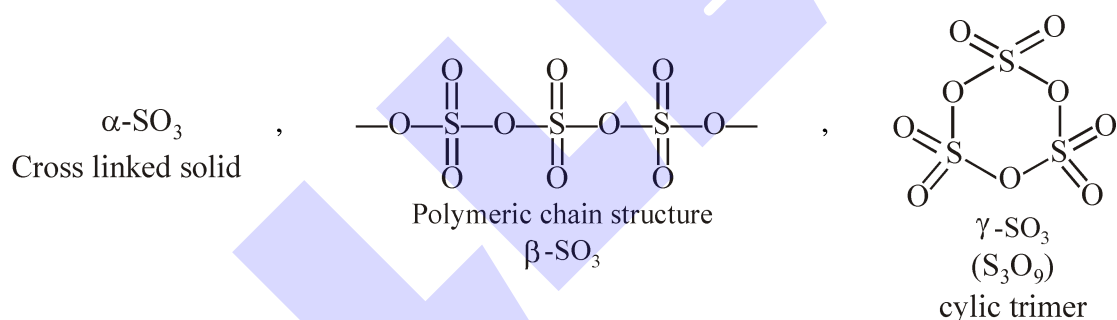


**Thermodynamic stability order :** Black phosphorus > Red phosphorus > White phosphorus

**Reactivity order :** Black phosphorus < Red phosphorus < White phosphorus

### ALLOTROPIC FORM OF $\text{SO}_3$

$\text{SO}_3$  have three allotropic forms  $\alpha\text{-SO}_3$ ,  $\beta\text{-SO}_3$  and  $\gamma\text{-SO}_3$



**NOTE:-** Si, Ge and Sn have a diamond type of structure, though Sn exists as a metallic form. Pb exists only in the metallic form. Ge is unusual because the liquid expands when it forms the solid. This property is unique to Ga, Ge and Bi.

